Computational Supplement:
Frame covariance of CMBR lensing in the exact $1+3$ covariant radiation hierarchy

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v1.3.0-dev — the derived recombination grid, the structural audit, the acoustic modes against the antecedent's own $1+3$ acoustic equations, and the localisation of the frame discrepancy (§8)
Previous release: v1.2.0, the full linear spectrum with the acoustic peaks in both frames. First public analytical reproducibility release: v1.0.0, 5 October 2026
20 October 2026

Abstract

This supplement is the analytical companion to *Frame covariance of CMBR lensing in the exact* $1+3$ *covariant radiation hierarchy* (Gebbie, Paper 1, University of Cape Town, 2026, `arXiv:astro-ph/9912072v4`, doi:10.48550/arXiv.astro-ph/9912072, submitted to *General Relativity and Gravitation* 2026-10-16, CC BY 4.0). **The version suffix is not optional**: a bare `astro-ph/9912072` resolves to whatever is current and has carried three distinct documents — `v4` is Paper 1 and `v2` is the 1999 correction. The bundle it documents is an **independent Python reconstruction, from first** $1+3$ **covariant principles**, of the almost-Friedmann–Lemaître results of Gebbie, Dunsby & Ellis, *Annals of Physics* **282**, 321–394 (2000). It is written bottom-up: not a port of, and not a wrapper on, an existing Boltzmann code.

What the computations here are. Model-conditional reproducibility evidence for a *linear* reconstruction: that the antecedent's equations, re-derived and evaluated independently, produce what the antecedent says they produce, and that where they do not the disagreement is located and named. The record of those disagreements is itself a deliverable — fifteen findings against the antecedent and forty-eight against this bundle, each with the measurement that settles it.

What they are not. They are not a calibration against data, not a fit, and not a claim of a new physical effect; the 1999 claim of one was withdrawn, and nothing here reinstates it. They are not a validation of the nonlinear extension, which this release does not contain. And they are not an assertion of agreement with CAMB or CLASS: the distance to those codes is *measured and itemised by approximation* rather than minimised, and acceptance is gated on this bundle's own pipeline.

One limit ships stated rather than discovered. The recombination grid sets the primary source at η_* and nothing derives it, which is worth 1.2% — larger than every tolerance this release claims, and invisible to the acceptance gate because both of its routes share that grid.

Contents

1 Scientific scope **2**

2 Theory-to-computation map **2**

3 The acceptance spine **9**

4 Conventions and their limits **10**

5 The generic source derivatives **17**

6 Frames and specialisation **18**

7 The transport assembly **23**
7.1 The W-form . . . 24
7.2 The $\ell = 0$ matching row . . . 25
7.3 The κ' ledger . . . 26
7.4 What this section establishes, and what it does not . . . 27

8 The frame discrepancy: localisation and triage **27**
8.1 The statement being tested . . . 27
8.2 Triage A — sites the discrepancy cannot come from . . . 28
8.3 The two measurements that closed the upstream end . . . 28
8.4 Triage B — the one frame-sensitive site that remains . . . 29
8.5 Triage C — frame-blind sites . . . 30
8.6 Triage by conformal time . . . 30
8.7 What would change this account . . . 31
8.8 The repair, attempted five times, and withdrawn . . . 31
8.9 What one term does to the integrand . . . 31
8.10 What this section does not establish . . . 32

9 Comparison and its range **32**
9.1 Criterion 5b's tolerance . . . 32

10 Evidence **33**

11 Release progression **34**
11.1 v1.0.0 — the reconstruction, and whether it is faithful . . . 34
11.2 v1.1.0 — the recombination that makes peaks possible . . . 34
11.3 v1.2.0 — the spectrum with structure, and the audit of it . . . 34
11.4 The arc, in one sentence each . . . 44
11.5 v1.3.0 — the structural audit, and the discrepancy localised . . . 44

12 The scattering-strength expansion, re-derived **44**
12.1 The recursion is geometric . . . 45
12.2 Two consequences the closed form makes immediate . . . 45
12.3 What this costs the oscillator . . . 45
12.4 What appears at second order that does not appear at first . . . 46

13 The CDM-frame division of the integral solution, term by term **47**
13.1 Conventions, stated . . . 47
13.2 What has to hold . . . 48
13.3 The three shifts . . . 48
13.4 The shift of (102), and why ΔC_0 has no partner . . . 48
13.5 The k^2 is (90)'s, not a mixed convention . . . 49
13.6 Measured . . . 49
13.7 Why the antecedent never meets this . . . 49
13.8 Three repairs tested, and one of them is untested . . . 49
13.9 What is open . . . 50

14 The almost-EGS bounds **50**

15 Parameter and numerical evidence **51**
15.1 The three backgrounds, side by side 51

16 Numerical resolution and reproducibility **52**

17 Interpretive limits **53**

1 Scientific scope

This supplement records the analytical companion to *Frame covariance of CMBR lensing in the exact* $1+3$ *covariant radiation hierarchy*, Gebbie, Paper 1, University of Cape Town, 2026, `arXiv:astro-ph/9912072v4`, doi:10.48550/arXiv.astro-ph/9912072, submitted to *General Relativity and Gravitation* 2026-10-16, CC BY 4.0. **Cited versioned throughout**: the bare identifier has carried three documents and `v2` is the 1999 correction, not this paper. The bundle is an independent Python reconstruction, from first $1+3$ covariant principles, of the almost-Friedmann–Lemaître results of Gebbie, Dunsby & Ellis, *Annals of Physics* **282**, 321–394 (2000), `doi:10.1006/aphy.2000.6034`. It is archived at `doi:10.25375/uct.34069509`, live since v1.0.0. That is the **concept** DOI and always resolves to the latest version, so it is the one to cite.

It is written bottom-up: not a port of, and not a wrapper on, an existing Boltzmann code, and it inherits no other code's conventions. It reproduces; it does not argue. No new effect is claimed.

Criterion 5b compares against CAMB and CLASS. It does not port them. From v1.2.0 this bundle ships a `reference/` directory holding TT spectra generated by CAMB and CLASS, and that is the only form in which either code appears. **Nothing under `functions/` imports `camb` or `classy`, and nothing ever may**: a reconstruction that imported the codes it is checked against would not be an independent one, and a reader reproducing figure F1 should not need a Fortran toolchain. `scripts/make_reference_spectra.py` is the single caller, it is not a figure stage, and what enters the repository is **data with a stamp** — the spectra, the versions that produced them, every setting, and the derived parameters — never vendored source. Where a published result is carried on this side so that a comparison is like for like, it is taken from the literature and named, not read out of either code. **The external codes are a check on this reconstruction, never an input to it.**

The background expansion is *derived* from the $1+3$ covariant energy constraint rather than transcribed from a printed expansion law, so that the antecedent's own Eqs. (185) and (G.3) are checks on this bundle and not its sources. That ordering is what makes a reconstruction an audit, and it is load bearing: a port would have inherited the misprint recorded below.[1]

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2 Theory-to-computation map

Table 1 links each published equation the numerics implement to its provenance and to the `.py` file and object implementing it, together with the test that checks it. Table 2 gives the notation and the names the quantities carry in the code.

Both are *generated* from `config/implementation-register.toml` by `scripts/make_tables.py`, and `tests/test_register.py` resolves every module, object and test named in them. They cannot drift from the code: a rename breaks the build rather than quietly falsifying a table printed here.

The harmonic normalisation is $Q_{A_\ell} = (-k_{\rm phys})^{-\ell}\mathrm{D}_{\langle A_\ell\rangle}Q$. Both candidate conventions are run end to end; only this one reproduces the real, opposite-sign form of the external hierarchies.[2]

Nomenclature is kept strictly separate: τ_ℓ are **mode** coefficients and τ_{A_ℓ} are **multipole** coefficients. *Annals II* notes that the distinction is not made in Bardeen-variable treatments. Every released figure

[1] The Gauss constraint at (G.3) prints $+\frac{2}{3}\Theta^2$ where the companion thesis and the Cargèse lectures Eq. (55) carry $-\frac{2}{3}\Theta^2$. In the Friedmann limit the printed sign gives $H^2 = K/a^2 - (\mu+\Lambda)/3$, which for a flat section is $-(\mu+\Lambda)/3$ and so has no solution at positive energy density. A misprint, recorded; the derived background already carries the correct sign, and a test now pins it so that a later edit towards the printed form fails rather than passes.

[2] *Annals I* prints $(-1)^\ell$ at the corresponding step where its own definition gives i^ℓ. The covariant multipole τ_{A_ℓ} is invariant under the choice, so the misprint never produced a wrong result; the mode coefficients are not invariant, and the two statements are distinct. Both are asserted as tests. A misprint, recorded in `provenance/ANNALS-II-ANTECEDENT-AND-CORRECTIONS-v1.md`.

plots a multipole mean-square, which holds for general geometries, rather than a mode mean-square, which holds only for almost-Robertson–Walker ones.

Table 1: Equations implemented in the numerics, with their provenance and the code that implements them. Equation numbers are those of the *published* papers. Rows with no code are not yet implemented.

Quantity	**Source**	**Equation**	**File**	**Object**	**Checked by**
$\beta_\ell = O^{A_\ell} O_{A_\ell} = (\ell!)^2 2^\ell/(2\ell)!$	Annals I	(24)	`functions/harmonics/weights.py`	`beta`	`test_beta_module_matches_symbolic_exactly`
$\beta_\ell = \frac{\ell}{2\ell-1}\beta_{\ell-1}$, $\beta_{\ell+1} = \frac{\ell+1}{2\ell+1}\beta_\ell$	Annals I	(25)	`functions/harmonics/weights.py`	`beta`	`test_beta_recurrence_descending_general_ell`
$\Delta_\ell = 4\pi\beta_\ell/(2\ell+1)$	Annals I	(119)	`functions/harmonics/weights.py`	`delta_over_four_pi`	—
α_ℓ, defined by $\beta_\ell = \alpha_\ell^{-1}(2\ell+1)$	Annals II	App. F, p. 380	`functions/harmonics/weights.py`	`alpha`	`test_f4_is_direct_substitution_into_f3`
free-streaming weight $w_\ell = (\ell+1)^2/[(2\ell+3)(2\ell+1)]$	Annals II	(F.1), p. 379	`functions/harmonics/weights.py`	`free_streaming_weight`	`test_f1_weight_is_the_mode_recursion_weight_shifted`
covariant recursion $-\dot{\tau}_\ell = (k_{\rm com}/a)[w_\ell\tau_{\ell+1} - \tau_{\ell-1}]$	Annals II	(F.1), p. 379	`functions/harmonics/free_streaming.py`	`free_stream`	`test_the_three_normalisations_are_one_solution`
β-normalised recursion, conformal time	Annals II	(F.3), p. 380	`functions/harmonics/free_streaming.py`	`free_stream`	`test_the_three_normalisations_are_one_solution`
α-normalised recursion	Annals II	(F.4), p. 380	`functions/harmonics/free_streaming.py`	`free_stream`	`test_f4_variable_is_exactly_the_spherical_bessel_function`
free-streaming solution $\alpha_\ell^{-1}\tau_\ell = j_\ell(k_{\rm com}\Delta\eta)$	Annals II	App. F, p. 380	`functions/harmonics/free_streaming.py`	`analytic_projection`	`test_f4_variable_is_exactly_the_spherical_bessel_function`
$\dot{\Theta}_\ell = k[\frac{\ell}{2\ell-1}\Theta_{\ell-1} - \frac{\ell+1}{2\ell+3}\Theta_{\ell+1}]$	Hu & Sugiyama	(6), p. 2601	`functions/harmonics/external.py`	`WEIGHTS`	`test_external_hierarchies_match_the_bundle_at_machine_precision`
$\dot{F}_{\nu\ell} = \frac{k}{2\ell+1}[\ell F_{\nu(\ell-1)} - (\ell+1)F_{\nu(\ell+1)}]$	Ma & Bertschinger	(49), (50)	`functions/harmonics/external.py`	`WEIGHTS`	`test_external_hierarchies_match_the_bundle_at_machine_precision`
$\dot{\Delta}_{T\ell} = \frac{k}{2\ell+1}[\ell\Delta_{T(\ell-1)} - (\ell+1)\Delta_{T(\ell+1)}]$	Seljak & Zaldarriaga	(3d)	`functions/harmonics/external.py`	`WEIGHTS`	`test_external_hierarchies_match_the_bundle_at_machine_precision`

Quantity	Source	Equation	File	Object	Checked by
relative-sign control: the $c^2 = -1$ normalisation, run as a wiring check	Annals II	(F.1), p. 379	`functions/harmonics/free_streaming.py`	`free_stream`	`test_sign_control_fails_in_a_known_shape`
$\int_0^\infty z^{-m} j_\ell^2(z)\, dz$, corrected to carry $[(m/2)!]^2$	Annals II	(199), p. 372	`scripts/derive_ell_range.py`	`bessel_integral_corrected`	`test_corrected_bessel_identity_matches_quadrature`
$\ell_A = \pi\Delta\eta_*/r_s$, bounding criterion 5's range	Annals II	§8.3.2, p. 372	`scripts/derive_ell_range.py`	`acoustic_scale`	—
$(a'/a)^2 = H_0^2[\Omega_r a^{-2} + \Omega_m a^{-1} + \Omega_k + \Omega_\Lambda a^2]$	this work	the $1+3$ energy constraint	`functions/background/flrw.py`	`Background`	`test_the_gauss_constraint_sign_is_the_corrected_one_and_the_printed_one_has_no_solution`
$\mathcal{P}(k) = \mathcal{A}(k\eta_0)^{n-1}$	Annals II	(J.1)	`functions/spectra/transfer.py`	`power_spectrum_j1`	`test_j1_is_constant_at_n_equals_one_and_eta0_drops_out`
$\mathcal{P}(k) = \mathcal{B}k^n T^2(k)$	Annals II	(J.3)	`functions/spectra/transfer.py`	`power_spectrum_j3`	`test_j1_and_j3_differ_by_one_power_of_k`
$T(k) = [1 + (ak + (bk)^{3/2} + (ck)^2)^\nu]^{-1/\nu}$	Annals II	(J.5)	`functions/spectra/transfer.py`	`transfer_function`	`test_b10_the_printed_transfer_function_diverges_where_it_must_vanish`
$\dot\delta_\ell = -n_e\sigma_T\delta_\ell - ikT\{\frac{\ell}{2\ell-1}\delta_{\ell-1} + \frac{\ell+1}{2\ell+3}[1 - \frac{\ell(\ell+2)K}{k^2}]\delta_{\ell+1}\}$	Wilson	(8), p. 4	`functions/harmonics/external.py`	`integrate_wilson`	`test_wilson_reproduces_the_bundle_and_tests_the_phase_convention`
$x_e(z)$, equilibrium Saha	this work	standard; not in the antecedent	`functions/spectra/decoupling.py`	`saha_ionisation`	`test_saha_is_monotone_and_spans_the_full_range`
$\mathcal{V}(k,\eta) \simeq \kappa' e^{-\kappa}$	Annals II	§5.2	`functions/spectra/decoupling.py`	`RecombinationHistory`	`test_the_visibility_is_a_normalised_probability_density`
$\mathcal{D}(\eta_0,k) = \int d\eta\, \mathcal{V} e^{-(k/k_D)^2} \simeq e^{-(k/k_D)^2}$	Annals II	(101), §5.2	`functions/spectra/decoupling.py`	`RecombinationHistory`	`test_the_visibility_is_a_normalised_probability_density`
the integral solution with $e^{-\kappa}$ on gravitational and $\kappa' e^{-\kappa}$ on scattering sources	Annals II	(98), §5	`functions/spectra/decoupling.py`	`Weighting`	`test_the_two_weightings_are_distinct_objects`
the same solution with both weights replaced by $\mathcal{V}$	Annals II	(102), §5	`functions/spectra/decoupling.py`	`Weighting`	`test_the_visibility_weighting_cannot_carry_a_late_isw`

Quantity	Source	Equation	File	Object	Checked by
$\gamma \simeq \frac{k^2}{6\kappa'} \frac{R^2+\frac{4}{5}(1+R)}{(1+R)^2} = k^2/k_D^2$	Annals II	(168), §6	functions/spectra/decoupling.py	diffusion_scale	test_damping_suppresses_small_scales_and_leaves_large_ones
$-C_0 = \frac{5}{2}a\mathcal{B}_2$, $-C_1 = \frac{1}{k}(a\mathcal{B}_1+\kappa' v_B)$, $-C_2 = \frac{1}{k^2}a\mathcal{B}_2$	Annals II	(90), §5	—	—	—
$\tilde{\tau}_1' = -\frac{R'}{1+R}\tilde{\tau}_1 - \frac{k}{1+R}\delta\tilde{T} - k\Phi_A$; $\delta\tilde{T}' = -aH\Phi_A - \Phi_H' + \frac{k}{3}\tilde{\tau}_1$	Annals II	(152), (153), §7.1	functions/spectra/acoustic.py	solve_acoustic	test_the_pair_reduces_to_the_printed_oscillator_exactly
$\delta\tilde{T}'' + \frac{R'}{1+R}\delta\tilde{T}' + k^2 c_s^2 \delta\tilde{T} = -\Phi_H'' - \frac{R'}{1+R}\Phi_H' - \frac{k^2}{3}\Phi_A$	Annals II	(154), §7.1	functions/spectra/acoustic.py	solve_acoustic	test_the_pair_reduces_to_the_printed_oscillator_exactly
$c_s^2 = 1/(3(1+R))$, $R = 3\rho_M/4\rho_R$	Annals II	§6	functions/spectra/acoustic.py	sound_speed_squared	test_sound_speed_limits
$\delta\tilde{T} + \Phi_A \approx [\delta\tilde{T}(0) + (1+R)\Phi_A]\cos(kr_s) - R\Phi_A$	Annals II	(180), §7.1.2	functions/spectra/acoustic.py	sachs_wolfe_at_last_scattering	test_the_integration_converges_to_the_closed_form_180_as_r_is_held_constant
$[\delta\tilde{T} + \Phi_A](\eta_*) \approx \frac{1}{3}\Phi_A(0)(1+3R_*)\cos(kr_s^*) - R_*\Phi_A(0)$	Annals II	(181), §7.1.2	functions/spectra/acoustic.py	sachs_wolfe_at_last_scattering	test_b8_equation_181_sign_is_fixed_by_its_own_limit
$\frac{\beta_\ell \tau_\ell(\eta_0)}{2\ell+1} = S_P j_\ell(k\Delta\eta_*) + \int d\eta\,(S_{\rm DISW} + S_{\rm ISW}) j_\ell(k\Delta\eta)$	Annals II	(176), §7.1.1	functions/spectra/sources.py	temperature_transfer	test_176_with_an_impulsive_source_is_the_free_streaming_projection
$S_P = D(\eta_0, k)\,[(\tilde{\delta}_T + \Phi_A) + \kappa' v_B]$	Annals II	(177), §7.1.1	functions/spectra/sources.py	source_primary	test_176_with_an_impulsive_source_is_the_free_streaming_projection
$S_{\rm DISW} = V e^{-(k/k_D)^2}[\frac{1}{3}k\tilde{\tau}_1 + (\kappa'\tilde{v}_B' + \kappa''\tilde{v}_B)]$	Annals II	(178), §7.1.1	functions/spectra/sources.py	source_doppler	test_the_printed_doppler_bracket_equals_the_contracted_form
$S_{\rm ISW} = V e^{-(k/k_D)^2}[(\Phi_A' - \Phi_H') - aH(\tilde{\delta}_T + 3\Phi_A)]$	Annals II	(179), §7.1.1	functions/spectra/sources.py	source_integrated	test_b6_sign_alternative_is_not_a_no_op
$C_\ell = \frac{2}{\pi}\frac{\beta_\ell^2}{(2\ell+1)^2}\int \frac{dk}{k} k^3 \lvert\tau_\ell\rvert^2$	Annals II	(186), p. 368	functions/spectra/angular.py	cl_mode_route	test_the_harmonic_normalisation_cancels_out_of_186_exactly
$\langle \tau_{A_\ell} \tau^{A_\ell} \rangle = \frac{\beta_\ell}{2\pi^2}\int k^2 dk \lvert\tau_\ell\rvert^2$	Annals II	(187), p. 368	functions/spectra/angular.py	multipole_mean_square	test_188_is_delta_over_2l_plus_1_times_187

Quantity	Source	Equation	File	Object	Checked by
$C_\ell = \Delta_\ell (2\ell+1)^{-1} \langle \tau_{A_\ell} \tau^{A_\ell} \rangle$	Annals II	(188), p. 368	functions/spectra/angular.py	cl_covariant_route	test_the_two_routes_agree_to_machine_precision_on_real_input
$C_\ell^{\mathrm{SW}} = \frac{1}{2\pi} H_0^4 \Omega_0^{1.54} \int \frac{dk}{k^2} P(k) j_\ell^2(k\Delta\eta_*)$	Annals II	(198), p. 371	functions/spectra/sachs_wolfe.py	cl_sachs_wolfe_quadrature	test_the_analytic_reduction_matches_the_quadrature
large-scale closed form, $n = 1$	Annals II	(200), (201), p. 372	functions/spectra/sachs_wolfe.py	cl_large_scale_reduced	test_b5_printed_201_is_short_by_the_comoving_distance
closed-form k-integral kernel at complex m, $\int z^{-m} j_\ell^2(z) dz$	Annals II	(199), p. 372, corrected	functions/spectra/kernels.py	bessel_square_kernel	test_complex_kernel_reduces_to_the_real_identity
primary-term k-integral by power-law decomposition	Annals II	(176) primary term, (198)	functions/spectra/kernels.py	fftlog_primary_integral	test_fftlog_matches_quadrature_on_a_non_power_law_integrand
integrated terms, $\int d\eta\, S(k,\eta)\, j_\ell(k\Delta\eta)$	Annals II	(176), p. 365	functions/spectra/bessel_table.py	line_of_sight	test_line_of_sight_matches_direct_evaluation
truncation condition: four consecutive vanishing harmonics force $\sigma_{ab} = 0$	Annals II	App. E, (E.1), p. 378	—	—	—
C_0, C_1, C_2 from the rank-by-rank match of the integral solution	Annals II	(89)	functions/spectra/coefficients.py	match_coefficients	test_all_three_of_90_are_reproduced_by_the_match
$-C_0 \simeq +\frac{5}{2} a\mathcal{B}_2$, $-C_1 \simeq \frac{1}{k}(a\mathcal{B}_1 + \kappa' v_B)$, $-C_2 \simeq \frac{1}{k^2} a\mathcal{B}_2$	Annals II	(90)	functions/spectra/coefficients.py	c0_from_printed_89	test_b11_is_a_sign_in_89_not_in_90
$-\tilde{C}_1 \approx \kappa' \tilde{v}_B + (\delta\tilde{T} + \Phi_A)$	Annals II	(109)	functions/spectra/coefficients.py	newtonian_coefficients	test_the_newtonian_specialisation_is_equation_109
the slow-decoupling integral solution carried with a generic u^a: two boundary terms and one integrated term	Annals II	(102)	functions/spectra/integral_solution.py	integral_solution	test_the_specialisation_recovers_what_the_pipeline_computes
$(a\tilde{\mathcal{B}}_1)' \approx -a^2 H \tilde{\mathcal{B}}_1 + k(\Phi_A' - \Phi_H') - 2Hak\Phi_A + \frac{1}{3}k^2\tilde{\tau}_1$	Annals II	(106)	functions/spectra/coefficients.py	newtonian_c1_prime	test_t23_splining_the_amplitude_is_wrong_and_the_test_can_tell
$\dot{\sigma}_{\langle ab\rangle} + 2H\sigma_{\langle ab\rangle} - \mathcal{D}_{\langle a} A_{b\rangle} + E_{\langle ab\rangle} \simeq \frac{1}{2}\pi_{\langle ab\rangle}$, and its second derivative	Annals II	(G.12)	functions/spectra/coefficients.py	generic_shear_derivatives	test_the_rank_two_control_passes_symbolically

Quantity	Source	Equation	File	Object	Checked by
C_2, C_2' and C_2'' from $\mathcal{B}_2 = \sigma_{ab}$	Annals II	(90)	`functions/spectra/coefficients.py`	`c2_derivatives`	`test_no_numerical_derivative_survives_in_the_c2_chain`
$\sigma_{ab} = \mathcal{D}_{\langle a} v_{b\rangle}$ in the CDM frame, with $\dot{\sigma}$ and $\ddot{\sigma}$	Annals II	(31) + (G.12)	`functions/spectra/coefficients.py`	`cdm_frame_shear`	`test_B13_the_printed_n_t1_does_not_close_and_the_general_rule_does`
$C_0, C_1, C_2, C_1', C_2''$ in the CDM frame, $A_a = 0$	this work	(89) + (31)-(37) + (106) + (152)	`functions/spectra/coefficients.py`	`cdm_set`	`test_the_frame_shift_of_every_coefficient_is_the_closed_form_one`
$\tilde{\tau}_a = \tau_a - v_a$, as $\tilde{\tau}_1 \to \tilde{\tau}_1 - v/k$	Annals II	(37)	`functions/spectra/frame_transform.py`	`FrameShift`	`test_T27_the_dipole_carries_the_same_inverse_k_as_the_monopole`
$(\rho, p, \mathcal{H}')$ from the Ω's, consistent with $\dot{H} = -H^2 - (\rho + 3p)/6$	Annals II	Appendix G	`functions/spectra/coefficients.py`	`background_fluid`	`test_the_frame_shift_of_every_coefficient_is_the_closed_form_one`

Table 2: Notation, and the names the quantities carry in the code. The mode/multipole distinction is the one that matters most: *Annals II* notes that it is not made in Bardeen-variable treatments.

Symbol	In code	Meaning	Note
Φ_A, Φ_H	`phi_a, phi_h`	the two gauge-invariant potentials entering (177) and (179)	—
$\tilde{\delta}_T$	`delta_T`	covariant temperature perturbation	frame-dependent; $\tilde{\delta}_T + 3\Phi_A$ is the combination in dispute (B-6)
$\tilde{\tau}_1$	`tau_1`	the dipole, which supplies the Doppler term of (178)	—
$\tilde{v}_B$	`v_b`	baryon velocity	—
κ', κ''	`kappa_prime`	differential optical depth and its conformal-time derivative	supplied, never manufactured inside the source terms
V	`visibility`	visibility function, weighting the integrated sources	—
k_D	`damping_k`	diffusion damping scale, entering (178) and (179) as $e^{-(k/k_D)^2}$	—
aH	`conformal_hubble`	a'/a, the background combination (179) carries	derived from the $1+3$ constraint, not from a printed expansion law
S_P, $S_{\rm DISW}$, $S_{\rm ISW}$	`source_primary, source_doppler, source_integrated`	the primary, Doppler and integrated temperature sources of (177)–(179)	individually frame-dependent
$\tau(x, e)$	—	temperature anisotropy as a function of position and direction	—
τ_{A_ℓ}	—	**multipole** coefficients — the covariant objects	valid for general geometries; what every released figure plots
τ_ℓ	`tau, values`	**mode** coefficients	almost-Robertson–Walker only. Conflating these two is how a sign is lost
O^{A_ℓ}	—	$e^{\langle A_\ell \rangle}$, the PSTF basis on the sphere	Annals I (15)
Q_{A_ℓ}	—	harmonic basis tensor, $(-k_{\rm phys})^{-\ell} \mathrm{D}_{\langle A_\ell \rangle} Q$	the adopted normalisation; see criterion 3
β_ℓ	`beta`	$O^{A_\ell} O_{A_\ell}$	Hu & Sugiyama's normalisation
α_ℓ	`alpha`	$(2\ell+1)/\beta_\ell$	Ma & Bertschinger's and Seljak & Zaldarriaga's normalisation
$k_{\rm com}$	`k_com`	comoving wavenumber	**never a bare `k` in this codebase**
$k_{\rm phys}$	`k_phys`	physical wavenumber, $k_{\rm com}/a$	Appendix F's k is comoving; the conventions sheet's is physical
η	`eta`	conformal time, $dt = a\, d\eta$	—
$\Delta\eta_*$	`d_eta`	$\eta_0 - \eta_*$	—
j_ℓ	`spherical_jn`	spherical Bessel function	the free-streaming solution
u^a	—	the frame 4-velocity	carried generic; frames are specialisations made at the end
σ_{ab}	—	shear	σ is the shear and nothing else
A_a	—	acceleration, $\dot{u}_a$	—
Θ_ℓ	—	Hu & Sugiyama's multipole variable	β-normalised

3 The acceptance spine

Appendix F of the antecedent, pp. 379–380, Eqs. (F.1)–(F.4), links the covariant mode recursion to external treatments written in real coefficients. Reproducing that chain tests the harmonic normalisation

without introducing a representation, and it is where this bundle's independence lies.[3]

Five external treatments are read and **four** are drawn — Hu & Sugiyama Eq. (6), Wilson (1983) Eq. (8), Ma & Bertschinger Eqs. (49) and (50), and Seljak & Zaldarriaga Eq. (3d) — each transcribed from its own paper.[4] The fifth is Wilson & Silk (1981) Eq. (7), the earliest of them and likewise read from its own page: at $K = 0$ it *is* Wilson (1983) Eq. (8), so it is credited and not drawn, four panels carrying four independent numerical checks. Hu & Sugiyama and Wilson write the recursion in the β normalisation — the normalisation their Eq. (2) fixes — and the other two in the α normalisation, which is why Appendix F prints both (F.3) and (F.4). A basis phase leaking into the couplings would break at least one of the two matches.

Wilson is the strongest of the four, and not because of his weights, which are Hu & Sugiyama's. He writes the recursion in the *imaginary* convention, with both bracket terms positive under an overall $-ikT$, where the others carry opposite signs under a real k. The two are related by $\Theta_\ell = i^\ell \delta_\ell$, which turns one form into the other exactly. He is integrated complex and as printed, and the phase removed only afterwards; transforming the equation first would assume the identity under test. That the transformed solution is *real* to 10^{-10} is a separate assertion from that it agrees. Wilson therefore tests the phase convention — the subject of §2's footnote and of acceptance criterion 3 — from a source published before any of the others.[5]

4 Conventions and their limits

The same hierarchy is written four different ways in the literature this bundle checks against, and the differences are not cosmetic. Table 3 is the translation, generated from `config/implementation-register.toml` so that it cannot drift from the code, with the test that checks each row.

"RW", "FL" and "FLRW" are three different claims, and this bundle makes the middle one. The names are used interchangeably often enough that it is worth separating them once.

- **Robertson–Walker (RW)** is a statement about *geometry*: the spacetime is spatially homogeneous and isotropic. It says nothing about what drives the expansion, and it follows from symmetry alone — no field equation has been used.
- **Friedmann–Lemaître (FL)** is that geometry *with its dynamics*: the Einstein equations imposed on it, giving the Friedmann and Raychaudhuri equations and hence $a(\eta)$ for a given matter content. FL is RW *plus* the field equations.
- **FLRW** is the two together, and is the usual compound name.

Annals II and this bundle say **almost-Friedmann–Lemaître**, and the choice is not stylistic. The background here is integrated from the $1 + 3$ energy constraint rather than assumed, so the dynamical

[3] The comparison applies the $(2\ell + 1)^{-1}$ division that the text preceding (F.4) specifies and the display omits; (F.4) is otherwise correct as printed. A misprint, recorded.

[4] Ma & Bertschinger print the same recursion twice. Eqs. (49) and (50) are the *massless neutrino* hierarchy, synchronous and conformal Newtonian, and are collisionless; their Eqs. (63) and (64) are the photon hierarchy, carrying the identical recursion plus a Thomson term $-an_e\sigma_{\rm T}F_{\gamma\ell}$. Appendix F's (F.4) is a free-streaming recursion, so (49) and (50) are the comparison and (63) and (64) are not. Published and arXiv numbering agree for this paper, which is not the case for the Annals papers.

[5] Appendix F's citation pairs the name of one paper with the key of another. Both papers are real and both numbers are right for the paper they belong to: Wilson & Silk (1981) Eq. (7) *is* the $\ell > 2$ hierarchy, and Wilson (1983) Eq. (7) is that paper's quadrupole equation while its Eq. (8) is the hierarchy. The prose names Wilson & Silk, the key is Wilson alone and consistent across every citation, and the load-bearing $i^{-\ell}$ remark names Wilson alone — so the match and the phase remark are about Wilson (1983), and the number beside them is correct for the 1981 paper the prose names. One mismatched pairing, not two independent misprints; recorded, and it moves no result.

half is in use from the start: Eqs. (185) and (G.3) are **checks on this bundle rather than inputs to it**. A treatment that only assumed almost-RW would have the geometry and not the expansion history, and could not perform that check.

The distinction is also what the almost-EGS theorem is about, and Sec. 14 depends on reading it the right way round. Almost-isotropy of the measured radiation bounds the *geometry* — an almost-RW conclusion. Getting from there to almost-FL requires the field equations as a second step. Collapsing the two names collapses a two-step argument into an assertion, which is precisely the move that paper is written to avoid.

There are exactly two conventions, and they differ by $c^2 = -1$. Changing the harmonic normalisation $Q_{A_\ell} = \mu^{-\ell}\mathrm{D}_{\langle A_\ell\rangle}Q$ rescales the mode coefficients as $\tau_\ell \to c^\ell\tau_\ell$ with $c = \mu'/\mu$. A recursion $\tau'_\ell = k[A_\ell\tau_{\ell-1} + B_\ell\tau_{\ell+1}]$ then carries $(A_\ell/c,\, B_\ell c)$, so the ratio B/A picks up c^2. At $c = i$ an opposite-sign real bracket becomes a same-sign bracket under an overall i. Hu & Sugiyama, Ma & Bertschinger and Seljak & Zaldarriaga print the first; Wilson prints the second. That is the whole of the difference, and it is asserted symbolically rather than described.

What the match establishes. The ℓ-weights, which are convention-independent, are confirmed by five sources in two normalisations, four of them drawn and numerically independent. The bundle is shown to sit at the same c^2 as Hu & Sugiyama, Ma & Bertschinger and Seljak & Zaldarriaga, and Wilson is shown to sit at the other. *Both* Wilson papers are in the imaginary convention, which settles a standing hypothesis in the negative: no author published this recursion in both conventions, so the split is between authors and not within one.

One source fixes its own phase, and that matters. Ma & Bertschinger define their Legendre expansion with an explicit factor $(-i)^\ell(2\ell+1)$ and state why: it is "chosen to simplify the expansion of a plane wave", so that $\exp(-i\mathbf{k}\cdot\mathbf{x})$ has coefficients $F_\ell = j_\ell(kr)$. Their coefficients therefore sit at $(-i)^\ell$ from the plane wave *by their own definition*, are real, and satisfy the opposite-sign real recursion. For this source the phase is read off the paper rather than assumed, which is exactly the standing the next paragraph says three of the four lack.

What it does not establish, and this release says so. It does **not** establish that $Q_{A_\ell} = (-k_{\mathrm{phys}})^{-\ell}\mathrm{D}_{\langle A_\ell\rangle}Q$ is the right reading of *Annals I*'s definition. Rephasing *both sides* of a comparison by i^ℓ is a change of variable and cannot fail — measured at 6.2×10^{-15} either way — so a match constrains the convention only where the phase applied to the external source is fixed from outside the comparison. For two of the four, no phase is applied, and that is a choice rather than a derivation; Ma & Bertschinger and Wilson are the two that are fixed from outside the comparison.

The exception, and it is the useful one. *Annals II* states, in the line following its flat-case mode functions, that the covariant form "differs by a factor of $i^{-\ell}$ from Wilson since we are using plain mode functions instead of plane waves". That phase is specified by the antecedent, not chosen here. So Wilson's panel in Figure 1 is the one comparison carrying information about the convention, and it is checked in two independent ways: the transformed solution agrees with the reconstruction, *and* it is real to 10^{-10}, which the agreement alone would not show.

Acceptance criterion 3 was therefore **reopened** and restated to the claim the evidence supports. Settling it needs *Annals I*'s definition of Q_{A_ℓ}, not a better match. Nothing computed here depends on it: the covariant τ_{A_ℓ} is invariant under the choice, the mode coefficients are not, and every released figure plots the invariant quantity.

Table 3: How each treatment in the foundational literature writes the same hierarchy, and how this bundle checked it. The two conventions in the literature are related by $\tau_\ell \to c^\ell \tau_\ell$ with $c^2 = -1$ and nothing else. **Where the transformation applied is "none" that is a choice, not a derivation** — which is why acceptance criterion 3 was reopened; see §4.

Treatment	**Variable**	**Bracket weights**	**Phase convention**	**Transformation applied here**	**How it was checked**
Annals II (F.1), Appendix F	τ_ℓ	$\frac{(\ell+1)^2}{(2\ell+3)(2\ell+1)}$ on $\tau_{\ell+1}$, 1 on $\tau_{\ell-1}$	covariant; $Q_{A_\ell} = (-k_{\text{phys}})^{-\ell} \mathrm{D}_{\langle A_\ell \rangle} Q$	—, this is the reference	arithmetic on β_ℓ alone; no representation introduced
Annals II (F.3), Appendix F	$\beta_\ell \tau_\ell$	$\frac{\ell}{2\ell-1}$, $\frac{\ell+1}{2\ell+3}$	real, opposite-sign bracket	none	(F.1)→(F.3) by multiplying through by β_ℓ, as exact rationals
Annals II (F.4), Appendix F	$\alpha_\ell^{-1} \tau_\ell$	$\frac{\ell}{2\ell+1}$, $\frac{\ell+1}{2\ell+1}$	real, opposite-sign bracket	the $(2\ell+1)^{-1}$ the prose states and the display omits	**finding B-2**: printed (F.4) is not yet in the Ma & Bertschinger form
Hu & Sugiyama Eq. (6), p. 2601	Θ_ℓ	$\frac{\ell}{2\ell-1}$, $\frac{\ell+1}{2\ell+3}$	real, opposite-sign bracket	none — *a choice*, see Q1	integrated as printed from its own paper, never through Appendix F
Wilson Eq. (8), p. 4	δ_ℓ	$\frac{\ell}{2\ell-1}$, $\frac{\ell+1}{2\ell+3}$	**imaginary**; both bracket terms + under an overall $-ikT$	$\Theta_\ell = i^\ell \delta_\ell$ — **fixed by the antecedent**, which states the $i^{-\ell}$ itself	integrated *complex and as printed*; the phase removed only afterwards, and the result asserted real to 10^{-10} separately from that it agrees
Wilson & Silk Eq. (7), p. 15	δ_ℓ	$\frac{\ell}{2\ell-1}$, $\frac{\ell+1}{2\ell+3}$	**imaginary**, as Wilson 1983 — *not* the real form	none required: at $K = 0$ it is Wilson 1983 (8) exactly	read from its own page. The **earliest** of the five and a fifth source for the weights, but **not** a fifth numerical check — the same equation cannot check itself
Ma & Bertschinger Eqs. (49), (50)	$F_{\nu\ell}$	$\frac{\ell}{2\ell+1}$, $\frac{\ell+1}{2\ell+1}$	real, opposite-sign bracket	none — *a choice*, see Q1	integrated as printed from its own paper; synchronous and conformal Newtonian gauges

Treatment	**Variable**	**Bracket weights**	**Phase convention**	**Transformation applied here**	**How it was checked**
Seljak & Zaldarriaga Eq. (3d)	$\Delta_{T\ell}$	$\frac{\ell}{2\ell+1}$, $\frac{\ell+1}{2\ell+1}$	real, opposite-sign bracket	none — *a choice*, see Q1	integrated as printed; also the method CMBFAST implements, cited for criterion 5b

The second table, and what its two thesis columns are for. Table 3 translates between the treatments this bundle is *checked against*. Table 4 does something different: it writes each coupling of this reconstruction in the conventions of Maartens, Gebbie & Ellis (1999), Challinor (2000), the thesis in **both** of the conventions it uses, Beneke & Fidler (2010) and Pitrou (2009). Every cell is derived from a named line of the normative sheet, and no cell is blank: where a treatment has no established counterpart the cell says so.

Two things in it are worth reading rather than skimming. **The thesis takes two columns and they differ on exactly the kinematic rows** — its $\dot{\mathcal{T}}$ form moves every kinematic term to the right-hand side, flipping the overall sign against its K/Π form. Elsewhere the two agree and the table says so in as many words, because a difference in wording where there is none in substance is precisely how two conventions get conflated while appearing distinct. And **the Beneke & Fidler and Pitrou columns are mostly "not established"**: the sheet records both as unsettled at the level of hierarchy coefficients, Beneke & Fidler working in (ℓ, m) tetrad form and Pitrou in Poisson gauge, and neither has had a hierarchy coefficient reproduced here. That is the honest content of those columns and it is printed rather than left to inference.

Table 4: The couplings of this reconstruction written in each external convention. Every cell is derived from the line of `provenance/conventions.md` named in the *Sheet* column; no cell is blank, and a treatment with no established counterpart says so rather than being left to inference. **The thesis takes two columns and they differ on exactly the kinematic rows** — the $\dot{\mathcal{T}}$ form moves every kinematic term to the right-hand side, flipping its overall sign against the K/Π form. On the remaining rows the two agree, and the table says *as Th-K* rather than restating them: a difference in wording where there is none in substance is how two conventions get conflated while appearing distinct.

Coupling	**Sheet**	**MGE99**	**CL**	**Th, K/Π**	**Th, $\dot{\mathcal{T}}$**	**BF**	**P09**
Rank raising	`C4.2`	identity	identity in form; D_a unchanged, the sign sits in k	identity	as Th-K	not established: BF work in (ℓ, m) tetrad form, no coefficient reproduced	not established: Pitrou works in Poisson gauge, no hierarchy coefficient checked
Rank lowering	`C4.4`	identity	identity in form; CL (2.25) prints $+\mathrm{D}^b I_{bA_\ell}$, reproduced from MGE99 (71)	identity; Th App. D footnote to (D.44) at $\ell = 1$ agrees	as Th-K	not established	not established
Free-streaming mode recursion	`C4.5`	identity	identity in form; CL (2.25) free streaming reproduced from MGE99 (71)	identity	as Th-K	not established	not established
PSTF contraction on the sphere	`C6.1`	identity (one source: EMT 1983, via C6.4)	identity; CL expands in the same O^{A_ℓ}	identity, Th (1.20)	as Th-K	not established: BF use ${}_sY_{\ell m}$, not a PSTF basis	identity at the basis level: Pitrou (1.28) expands in n^{A_ℓ}
PSTF self-contraction β_ℓ	`C5.2`	identity	identity; CL absorbs it into Δ^C_ℓ, see the row below	identity, Th (1.23)	as Th-K	no counterpart: (ℓ, m) normalisation carries $\sqrt{4\pi/(2\ell+1)}$ instead	identity, Pitrou (1.29) in the equivalent $4\pi\ell!/(2\ell+1)!!$ form
Angular measure Δ_ℓ	`C5.5`	identity, MGE99 (47), same value	$\Delta^C_\ell = 4\pi(-2)^\ell(\ell!)^2/(2\ell+1)! = (-1)^\ell\Delta_\ell$, CL (2.23)	identity	as Th-K	no counterpart in this form	identity, Pitrou (1.29), verified symbolically

Coupling	**Sheet**	**MGE99**	**CL**	**Th, K/Π**	**Th, $\dot{\mathcal{T}}$**	**BF**	**P09**
Shear coupling at $\ell+2$	`C9, T-MGE and T-Th-T`	$-(\ell+2)$ as printed in (71), (88), (90), (99) — **defect E9**; $+(\ell+2)$ is correct, and MGE99 (89) at $\ell=2$ prints $+\frac{8}{15}\rho\sigma$, agreeing with the correction	$-\frac{8}{15}\delta_\ell^2 I\sigma$ gives $+(\ell+2)$, independently of MGE99	$-(\ell+2)$ in Th (2.73) — carries E9	$+(\ell+2)$ **on the RHS** in Th (2.84): every kinematic term flips overall sign against (2.73) because it is moved across. **This row is where the two thesis columns differ**	not established: BF put the observer at rest, so they carry no shear coupling	not established
Acceleration coupling at $\ell\pm1$	`C9, T-CL and T-Th-T`	identity	$A_a \to -A_a$ under $g \to -g$ at fixed u^a (CL footnote to §2), the signature difference and nothing more	identity	overall sign flipped with the rest of the kinematic terms — **differs from Th-K**	not established: observer at rest	not established

5 The generic source derivatives

*Decision **S26**. Written from `provenance/GENERIC-SOURCE-DERIVATIVES-v1.md`, which carries the working and the open items.*

What (102) needs, and what the paper never wrote. (102)'s integrand is $C_0 - C_1' + C_2''$ and its boundary terms carry $C_2'(\eta_*)$. With (90),

$$-C_0 \simeq +\tfrac{5}{2}a\mathcal{B}_2, \qquad -C_1 \simeq \tfrac{1}{k}(a\mathcal{B}_1 + \kappa' v_B), \qquad -C_2 \simeq \tfrac{1}{k^2}a\mathcal{B}_2,$$

Two entries of that display are corrected in §7. The $+\frac{5}{2}$ on C_0 is α_2, the rank-*two* normalisation, applied to a monopole row where $\alpha_0 = 1$ belongs — finding **T-104**, a factor of $\frac{15}{2}$ — and the $\kappa' v_B$ inside C_1 is a bare κ' in a *coefficient*, which §7.3 forbids and **T-101** measures on the released arrays. Both are stated here at the point of use so a reader of this display does not meet the correction four hundred lines later.

so the solution needs $(a\mathcal{B}_1)'$, $(a\mathcal{B}_2)'$ and $(a\mathcal{B}_2)''$ **for arbitrary** u^a. *Annals II* prints these only after the Newtonian specialisation has removed the shear, which is exactly the specialisation this bundle declines to make early. The derivations are therefore assembled here, and this section records what is closed, what is assembled, and what is open.

The three sources, by rank, with no frame chosen. From §4's integrated Boltzmann equation $\dot\tau + e^a \mathrm{D}_a\tau + \mathcal{B} \simeq C[x,e]$,

$$\mathcal{B} = -\tfrac{1}{3}\mathrm{D}^a\tau_a + (\mathrm{D}_a \ln T + A_a)e^a + \sigma_{ab}e^a e^b,$$

giving $\mathcal{B}_0 = -\frac{1}{3}\mathrm{D}^a\tau_a$, $\mathcal{B}_1 = \mathrm{D}_a \ln T + A_a$ and $\mathcal{B}_2 = \sigma_{ab}$. **All three vanish in the background**, so all three are gauge-invariant at first order by Stewart–Walker — which is the paper's own *"note that these are CGI with respect to u^a"*.

The scale-factor rule, and why the footnote to (105) carries $2H$. A rank-n PSTF source's natural variable carries a^n: the paper writes $(a\tilde{\mathcal{B}}_1)'$ at (106) and $(a^2\sigma_{ab})^{\cdot}$ in Appendix G. So

$$\dot{\mathcal{B}}_1 + 2H\mathcal{B}_1 = \frac{1}{a^2}(a^2\mathcal{B}_1)^{\cdot},$$

and the rank-two object built from a rank-one source, $\mathrm{D}_{\langle a}\dot{\mathcal{B}}_{b\rangle}$, carries the rank-two power. **It is not a commutator correction bolted on; it is the rank showing up in the measure.** That is conventions row **C3.1a** read in the direction nobody had needed, and it is checkable: under C3.1, C3.2 and C4.2 the mode functions are time-independent at every rank, so *all* the H's come from the a^n and none from the Q's — which **predicts (106)'s $2H$ before (106) is consulted**. Implementing it the other way, by splining a stored amplitude, is wrong by a median of 100%: finding **T-23**, and the reason `integral_solution` takes every derivative as an input rather than differentiating one.

Rank 2: closed. $\dot{\mathcal{B}}_2$ *is* in the paper generically, in Appendix G, and closes to second order **with an exact symbolic control** that fails when the rank-two term is removed — so the check is one that can fail, which S27's C rung requires and finding **T-24** exists because an earlier one could not.

Rank 1: assembled, and carrying one term more than (105) prints. $\dot{\mathcal{B}}_1$ is assembled from printed equations. Three of its four terms close exactly against (105)/(106) — $\mathcal{B}_0$, $\dot\Phi_A$ and $-2H\Phi_A$, the last only because both of its contributions are present — and substituting the expansion–potential relation of §*newtg* reproduces every term (105) prints. The generic form then carries **one extra**:

$$(\star) \qquad -H\,\mathrm{D}_{\langle a}\tilde{\mathrm{D}}_{b\rangle}\ln T.$$

This is the temperature-gradient half of $-H\tilde{\mathcal{B}}_a$, whose acceleration half *does* survive into (105). **Two readings, and this bundle chooses neither.** It may be the small-scale approximation — §6 drops all terms of $O(\epsilon\epsilon_H)$ and $(\star)$ has that shape — except that so does $-2H\mathrm{D}_{\langle a}\mathrm{D}_{b\rangle}\Phi_A$, which is kept. Or it may be an omission, in which case (105), (106) and what descends from them are short a term of the same order as ones they keep. It is first order and it does not obviously vanish, which is enough to make the question real; settling it needs §2.5's $O(\epsilon\epsilon_H)$ bookkeeping read against each term of (105), not a judgement about which is negligible. **Logged as open, not filed as a finding against the paper.**

Rank 0: not needed. Withdrawn 2026-10-21. This paragraph read: *"$\dot{\mathcal{B}}_0$ is not required by (102) at all — C_0 carries $\mathcal{B}_2$, not $\mathcal{B}_0$."* **It is wrong, and the reason is §7.2:** C_0 carries *both*, because (89)'s $a\mathcal{B}_0\delta_{\ell 0}$ row was dropped. Finding **T-105**. The earlier plan this paragraph overruled was right.

Status. The derivation is complete for what the integral solution needs. What remains open is $(\star)$ and the dropped $a\mathcal{B}_0\delta_{\ell 0}$ row of (89) — **both questions about the antecedent rather than gaps in this assembly**, and both recorded rather than resolved by assumption.

6 Frames and specialisation

"Frame" is being asked to do two jobs, and they are different objects. The word is used in this literature for something specified at a single event and for something specified throughout a region, and the two are not interchangeable. The PI's enumeration, recorded here because the rest of this section depends on it:

Object	What is specified	Where
Instantaneous observer rest frame	A unit timelike vector $u^a(p)$, determining a local time direction and an orthogonal rest space	At an event p
Local orthonormal frame	A tetrad $e_{\hat{\alpha}}{}^a(p)$, with $e_{\hat{0}}{}^a = u^a$, also choosing three spatial axes	At an event
Observer worldline	A timelike curve with tangent u^a	Along one observer's history
Threading	A smooth timelike congruence, represented by a unit tangent field $u^a(x)$	Throughout a spacetime region
Adapted orthonormal frame field	A tetrad field whose timelike leg follows the congruence	Throughout that region

Nothing in this bundle is a tetrad. No calculation here chooses spatial axes: every quantity is a PSTF tensor built from u^a and $h_{ab} = g_{ab} + u_a u_b$, so rows two and five are named only to be set aside. What the bundle actually uses is rows one and four, and it uses both:

- the **receiving observer's frame** — a u^a at one event, reception at η_0 — which is what the *measurement* is referred to, and in particular what the dipole is a statement about;
- the **CDM threading** and the **Newtonian threading** — unit tangent fields $u^a(x)$ on the whole region between last scattering and reception — which is what the *propagation* integral is carried along, and what σ_{ab}, A_a and q_a are properties of.

The general transformations (31)–(37) relate two *threadings*; the specialisations (40)–(45) stipulate one. $\tilde{\sigma}_{ab} = 0$ and $\tilde{A}_a = 0$ are conditions on a congruence throughout a region and are not statements about any single observer, which is why neither can be imposed "at the observer" and why the Newtonian one is admissible only in restricted cases. Figure F7's two panels are exactly this pair kept apart: the $\ell \geq 2$ panel compares two *threadings* carrying the same observable, and the dipole panel is where the *receiving*

observer's frame enters, because the dipole is the one multipole that is a statement about u^a at reception rather than about the congruence that brought the radiation there.

The words this bundle uses, and why none of them is retired. *Threading* names the congruence **as an object**. *Receiving observer's frame* names the local u^a **at reception**. *Frame* continues to name a **choice of observers** and, following *Annals II* §4.1's own usage, each named choice — the energy frame, the zero-acceleration (CDM) frame, the Newtonian frame. Every one of those is a threading in the first sense: the words sit at different levels and are not competing. Decision **S29a**.

No identifier is renamed. The code says `"cdm"` and `"newtonian"`, and *Annals II* prints "frame" for exactly these objects, so the code agrees with the source it reconstructs. The rule that a word naming two things should have one of them renamed is satisfied here by **naming the two objects**, which is what the table above does. And the rename operation is itself the hazard: a blanket substitution in this bundle inverted the meaning of diagnostic D7, whose subject is the CDM *threading* measured against the **total-energy** frame — the one place both senses appear in a single sentence, and therefore precisely what a substitution cannot see. Finding **T-33**. **A rename a regular expression can perform is a rename whose exceptions nothing will catch.**

Specialisation. The solution is carried with a **generic** u^a. The explicit frames appear as choices made at the end, not assumed at the start: the Newtonian frame $\tilde{\sigma}_{ab} = 0$, which is the specialisation *Annals II* itself makes at p. 365 in order to remove the shear terms, and the **CDM frame** $\tilde{A}_a = 0$. Carrying the generic frame and specialising late is the difference the $1 + 3$ covariant approach buys; a gauge-fixed treatment fixes the frame before the hierarchy is solved and cannot exhibit it.

"CDM" names two different things, and this supplement keeps them apart. *CDM the model* is a cosmology — flat, matter dominated, no Λ — and it is one of the two models this release delivers a spectrum for. *The CDM frame* is a choice of frame u^a: the congruence in which the cold dark matter is at rest. The two are independent. The ΛCDM *model* is computed in the same *frame*, and a change of model is not a change of frame. In the code the models are named `CDM_MODEL` and `LCDM_MODEL` and the frames `"cdm"` and `"newtonian"`, so the two cannot be confused at a call site.

The CDM frame is *not* the total-energy frame, and an earlier version of this supplement said it was. The definition used here is exact in general relativity and needs no approximation: cold dark matter is **pressureless and collisionless**, so it is acted on by no non-gravitational force and its worldlines are **geodesics**. The CDM frame is therefore the frame with $\tilde{A}_a = 0$, which is what *Annals II* §4.1 calls the *zero acceleration frame*, listing it as the second of three useful choices — the first being the **total-energy frame** $q_a = 0$ and the third the Newtonian frame. They are **three stipulations, not two**, and the paper gives the reason this calculation wants the second: it is "geodesic right through tight-coupling, slow decoupling and into the free-streaming era", whereas the energy frame is recommended there for *two* coupled species and this is a three-species problem that crosses decoupling.

The two agree only in the limit of matter domination. Setting $q_a = 0$ in a mixture gives $\rho_C v_C^a + \frac{4}{3}\rho_R v_R^a + \ldots = 0$, so in the CDM rest frame the total flux is *not* zero — the baryons and the photons are moving — and the energy frame tilts away by $(\rho_b + \frac{4}{3}\rho_R)/\rho_C$. Measured in diagnostic D7, that is **0.24** at last scattering for this release's background and **0.49** for the stamped Planck column, with the baryons carrying 37% of it. It is largest exactly where the acoustic physics is, and falls afterwards on both counts, since ρ_R/ρ_C decreases and the baryons stop moving with the photons once they decouple. Recorded as finding **T-29**; the amendment is decision **S8a**.

The one-line reconciliation, and it is the PI's. The CDM frame is the energy frame *of the CDM component*. For pressureless dust $T^C_{ab} = \rho_C u^C_a u^C_b$, so its flux relative to a frame $u^a = u^a_C + v^a$ is

$q_a^C = \rho_C v_a$ to first order, and $q_a^C = 0$ gives $v_a = 0$ — the CDM at rest, exactly. So the two conditions are the same statement about one species, and the conflation this supplement previously carried was the dropped qualifier: "energy frame" unqualified means the *total*, and the total includes the baryons and the photons, which are moving. Saying *of what* settles it.

The frames, and what each one fixes. *Annals II* §4.1 enumerates four (its sentence says "three"); the baryon and radiation rows below are frames the paper uses elsewhere and are included because they are the ones a reader is most likely to reach for next.

Threading	Defining condition	What it establishes
CDM *(used here)*	$u^a = u_C^a$, equivalently $\tilde{A}_a = 0$	CDM is at rest; its congruence is geodesic in the conserved-dust model, which is exact rather than approximate. *Annals II* calls it the *zero acceleration frame*.
Total energy	$q_a^{\text{total}} = 0$	Total momentum density vanishes; **individual species can still move**. Agrees with the CDM frame only to $(\rho_b + \frac{4}{3}\rho_R)/\rho_C$.
Baryon	$u^a = u_B^a$	Baryons are at rest; pressure and scattering can accelerate them, so this frame is **not** geodesic.
Radiation energy	$q_a^\gamma = 0$	The radiation brightness dipole vanishes; **higher multipoles can remain**.
Newtonian *(used here)*	$\sigma_{ab} = 0 = \omega_a$, where admissible	Useful for the scalar almost-FLRW comparison and for the Bardeen link; generally accelerated. Admissible only in restricted cases.
Constant expansion	$\mathrm{D}_a\Theta = 0$, where admissible	Removes the spatial expansion gradient relative to that frame; sometimes useful on small scales. Not used here.

Reading down the second column is the point: **each row fixes a different thing, and no two of them are the same condition**. A frame is chosen and then named, not inferred from what it is called.

One of these frames contains observers and the other does not. This is the reason the distinction is worth the space, and it is where the $1+3$ covariant account differs from a metric one in substance rather than in notation.

The **CDM frame is a congruence of observers**. It is defined by a property of the matter — pressureless, collisionless, therefore geodesic — and the definition is exact in general relativity, not a first-order statement. We observe from inside it: a detector in a bound structure is comoving with the matter to the accuracy linear theory works at, and after decoupling the baryons have fallen into the same flow as the cold dark matter, so the baryon and CDM frames coincide at η_0 where the measurement is made. *It is "us".*

The **Newtonian frame is not**. $\tilde{\sigma}_{ab} = 0$ is a condition imposed to simplify, not a property any matter has, and *Annals II* §4.1 says so in terms: the frame "is only consistent in restricted cases" and is "particularly useful in making comparisons with much of the analytic literature". It is the covariant counterpart of what conformal Newtonian gauge does in a metric treatment — admissible about almost-FLRW, and a device.

Why this costs nothing at first order and will not stay free. τ_{A_ℓ} for $\ell \geq 2$ is frame-invariant at first order, so the *observable* does not care which frame carried it there, and a calculation performed in the Newtonian frame returns what an observer in the CDM frame measures. That is the whole licence for the Bardeen-variable route, and it is a first-order licence.

At $O(\epsilon^2\ell)$ it lapses. The couplings v1.5.0 adds are products of two first-order quantities, and a product of two frame-dependent quantities is not frame-invariant just because each factor's leading

frame-dependence cancels. Then the frame has to be the one the observer is actually in, and that is the CDM frame, not a device.

This is what figure F5 is for. Its third panel is the amount of machinery that is frame-dependent — in the Newtonian frame one rank carries everything, in the CDM frame all three are occupied — and its first two panels are the first-order invariance that currently makes that difference harmless. The figure shows both at once, which is why it is the right figure to carry into v1.5.0: it is a picture of exactly what stops cancelling.

What the external codes use, and why criterion 5b has to check it. Criterion 5b compares this bundle against CAMB and CLASS. That comparison is only meaningful if the frame each code evolves in is known, because a quantity compared across two frames is not the same quantity. The correspondence below is **verified where it could be, and recorded at its real strength where it could not**; verifying it is a **precondition of 5b**, not an afterthought.

CLASS is verified from its own source, at `classy v3.4.1`. `perturbations.c` carries the comment *"cdm velocity vanishes in the synchronous gauge"*, and `index_pt_theta_cdm` is allocated as an evolved variable only when `gauge == newtonian`: in synchronous gauge θ_C is not evolved because it is zero. The decisive line is CLASS's own error message — *"In the synchronous gauge, it is not self-consistent to assume no CDM: the latter is used to define the initial timelike hypersurface"* — which says, in the code's own words, that **its synchronous gauge is realised by the CDM congruence**. That is the CDM threading of the table above, $u^a = u^a_C$. Both `gauge = synchronous` and `gauge = newtonian` were run and computed.

CAMB is corroborated but not verified to the same standard. Its Python layer exposes `v_newtonian_cdm` and `v_newtonian_baryon` as *derived outputs*, notes that *"L=1 dipoles are Newtonian Gauge"*, and carries an explicit `frame` argument for sources that are not frame-invariant — which is what a code evolving in one frame and converting on output looks like. That is corroboration from an interface, not a reading of the evolution equations, and the decisive statement lives in CAMB's own covariant formulation paper rather than in anything this bundle inspected. It is recorded at that strength and no higher.

Code	**Gauge used for evolution**	**Corresponding $1+3$ frame**
CLASS *(default)*	Synchronous, with CDM velocity divergence $\theta_C = 0$	CDM-comoving: $u^a = u^a_C$, $A_a = 0$; shear generally nonzero.
CLASS *(optional)*	Conformal Newtonian / longitudinal, selected with `gauge = newtonian`	Newtonian: scalar shear and vorticity vanish; acceleration generally nonzero.
CAMB	Covariant equations evaluated in the CDM frame, equivalent to CDM-comoving synchronous gauge	CDM-comoving: $v_{_C} = 0$, $A_a = 0$; shear generally nonzero.
Bolt.jl	Conformal Newtonian gauge, implemented as `BasicNewtonian`	Newtonian: CDM carries an explicitly evolved peculiar velocity relative to this frame.

Two things follow if the table holds. First, **CAMB and CLASS's default both evolve in the frame this bundle uses**, which is the convenient case: 5b compares like with like without a frame transformation in between. Second, **CLASS's `newtonian` option and Bolt.jl evolve in the other one**, so a comparison against either is a comparison across frames and must either transform or restrict itself to the frame-invariant $\ell \geq 2$ multipoles. Figure F5 is the measurement of what that costs: the totals agree and the decompositions do not.

That CAMB is described as using covariant equations in the CDM frame is the part most worth checking rather than assuming, because if true it makes the comparison closer than a gauge-versus-frame translation would suggest, and a comparison that is closer than it should be is the kind this bundle has already been wrong about once.

For readers coming from the metric approach: a frame is not a gauge. This is the distinction most easily lost in translation, and it is worth stating plainly because the two freedoms are often fixed together in a metric-based treatment and so look like one freedom.

- A **gauge** is a choice of coordinates — equivalently a choice of correspondence between the perturbed spacetime and the background. In the $1+3$ covariant approach it is **already dealt with**: every variable used here vanishes in the background, so by the Stewart–Walker lemma it is gauge-invariant by construction. There is no residual gauge mode to fix and none to worry about.
- A **frame** is a choice of the timelike congruence u^a along which the fluid and radiation variables are defined. It is a *physical* choice — which observers — and it survives after the gauge question has been answered. It is the only freedom left, and it is the one this release exhibits.

So the familiar correspondences are correspondences and not identities. Synchronous gauge is usually *completed* by frame along the CDM, and conformal Newtonian gauge is usually *completed* by frame along a shear-free congruence; in each case the gauge condition fixes coordinates and a separate stipulation fixes the observers. A statement that is frame-dependent here will appear in a metric treatment as a statement that depends on which gauge was chosen, and the two dependences are not the same dependence. That is why figure F5 can show the *total* common to two frames while the *decomposition* differs: the split into ranks is frame-dependent and the observable is not, and neither fact is about coordinates.

This is also what fixes the status of the next release. Because v1.0.0 is the general case, **v1.5.0's restricted case is a recovery and a check** — it must return v1.0.0's numbers, and that requirement is the regression test rather than a new result.

The covariance claim is established symbolically, and that is the stronger form. Acceptance criterion 2 asks that a change of frame $u^a \to \tilde{u}^a = u^a + v^a$ return the rank-one bracket unchanged up to $\dot{v}_a$. It does, *identically* in exact arithmetic rather than to a tolerance, and the cancellation is exhibited rather than quoted:

- $\tilde{\mathrm{D}}_{\underline{a}} \ln T = \mathrm{D}_a \ln T - H v_a$, the extra term arising because the projector moves with the frame and $\overline{\ln T} = -H$ in the background;
- $\tilde{A}_a = A_a + \dot{v}_a + v^b \nabla_b u_a$, where $v^b \nabla_b u_a$ is formed *in full* — Hv_a together with $\sigma_{ab} v^b$ and $\epsilon_{abc} v^b \omega^c$ — and the latter two leave at $O(\varepsilon^2)$ by series expansion rather than by hand;
- so $\tilde{\mathcal{B}}_1 = \mathcal{B}_1 + \dot{v}_a$ with **no shear and no** Hv. It is the *same* H twice, which is why the cancellation is structural and not a coincidence of two fitted coefficients.

The curvature half. The deflection depends only on $\Phi_A - \Phi_H$, whose trace-free screen-projected Hessian is the electric Weyl tensor, $E_{ab} \simeq \frac{1}{2}\mathrm{D}_{\langle a}\mathrm{D}_{b\rangle}(\Phi_A - \Phi_H)$. Since E_{ab} vanishes in the background, the Stewart–Walker lemma makes it frame-invariant at first order, and therefore so is $\Phi_A - \Phi_H$. Both potentials shift by the same $(v' + \mathcal{H}v)/k_{\mathrm{com}}$, and the residual $\tilde{E}_{ab} - E_{ab}$ is identically zero — **with a negative control**: a rule leaving Φ_H fixed, or flipping its sign, leaves a residual proportional to $(v' + \mathcal{H}v)$ and fails. The invariance is therefore forced rather than chosen to make a residual vanish. Only the dipole moves, $\tilde{\tau}_a \simeq \tau_a - v_a$; every τ_{A_ℓ} with $\ell \geq 2$ is unchanged, which is the standard any frame dependence surviving at $\ell \geq 2$ must meet: it is an artefact of the reduction, not physics.

The energy frame is derived, not stipulated. Imposing $\tilde{A}_a = 0$ gives $v' + \mathcal{H}v = -k_{\mathrm{com}}\Phi_A$ — the Euler equation of a pressureless geodesic fluid, which is what the cold dark matter obeys. The normalisation is corroborated independently: in the baryon-dominated limit $R \to \infty$, Eq. (152) reduces to that same equation, so the photon dipole and the frame velocity carry one normalisation.

What this release does *not* do, stated because the earlier draft of this section claimed it. It does not compute an assembled spectrum in two frames and draw the difference. Eqs. (177)–(179) are **Newtonian-frame expressions**: they descend from (106), which the corrections record flags in those words, and the Newtonian frame carries its whole $O(\ell)$ source in the acceleration pair while the CDM frame carries its whole source in the shear pair. Transforming the four input fields and re-evaluating the same equations therefore deletes the coupling that was carrying the source without supplying the one that replaces it: measured, the transfer function moves by a factor of about five, and under the opacity weighting the source collapses to two parts in a thousand. That is not a frame dependence of the sky. The figures which would draw it, and the convergence figure which would test footnote 29 on p. 365 in both frames, require the generic-u^a hierarchy and are deferred with it.

One coefficient is changed for criterion 5b, and only there. The photon quadrupole's Thomson collision term carries $\Pi = \Theta_2 + E_0 + E_2$, so polarisation feeds back into the temperature hierarchy and strengthens the diffusion damping. CAMB and CLASS both include this **structurally**: it is not an output option, and neither can be asked to leave it out — CLASS fails outright in `perturbations_solve` below `l_max_pol_g = 2`, because Π needs E_2, and CAMB exposes no switch for it. Truncating CLASS's polarisation hierarchy from 10 to 4 moves the acoustic band by 0.043%, which bounds the hierarchy's *depth* and not the *feedback*.

The only way to compare like for like is therefore to carry the polarised coefficient here, for that comparison alone. In (dif-damp)'s quadrupole term *Annals II* prints 4/5; Hu & Sugiyama's unpolarised form is 8/9; with polarisation it is 16/15. Measured on `PLANCK_MODEL`:

	coefficient	$k_D(\eta_*)$	ratio
Annals II (dif-damp), as printed	4/5	**619.9**	1.000
Hu & Sugiyama, unpolarised	8/9	588.1	0.949
with polarisation	16/15	**536.9**	**0.866**

a 13.4% reduction in k_D, which takes $e^{-(k/k_D)^2}$ from 0.660 to 0.574 at $k = 400$ and from 0.189 to 0.109 at $k = 800$. It is one of the dominant terms in criterion 5b's residual.

4/5 **remains the default and the released value.** Every figure, diagnostic and released number in this bundle uses it, because the bundle reproduces and 4/5 is what the page prints. 16/15 is reached only through an explicit argument, by criterion 5b. Decision **S31**, finding **T-39**. *Annals II*'s 4/5 matches neither standard coefficient, which is recorded as a **question** — it may be a convention in how R and the factor of 6 are distributed — and not as a correction.

7 The transport assembly

The reader who wants only the result: §7.1 replaces the integration by parts, §7.2 replaces (90)'s C_0, and §7.3 is the rule that makes both of them checkable before they are implemented.

Three questions about the line-of-sight solution are one question, and this section answers it once. They are: which weight carries which source; what the $\ell = 0$ matching row of (89) should be; and why every repair attempted on (102) so far has moved a weight without moving the term that dominates the integrand. The common answer is that **(102)'s integration by parts was taken onto the coefficients**, and that the assembly in which that step never arises is the one obtained by multiplying the transport equation by its own integrating factor.

Figure 1 is the order this section and §8 are written in. It is the photon's history and not a table of contents: each open question is drawn at the conformal time it acts at, which is what separates the questions that a frame instrument can see from the questions it cannot.

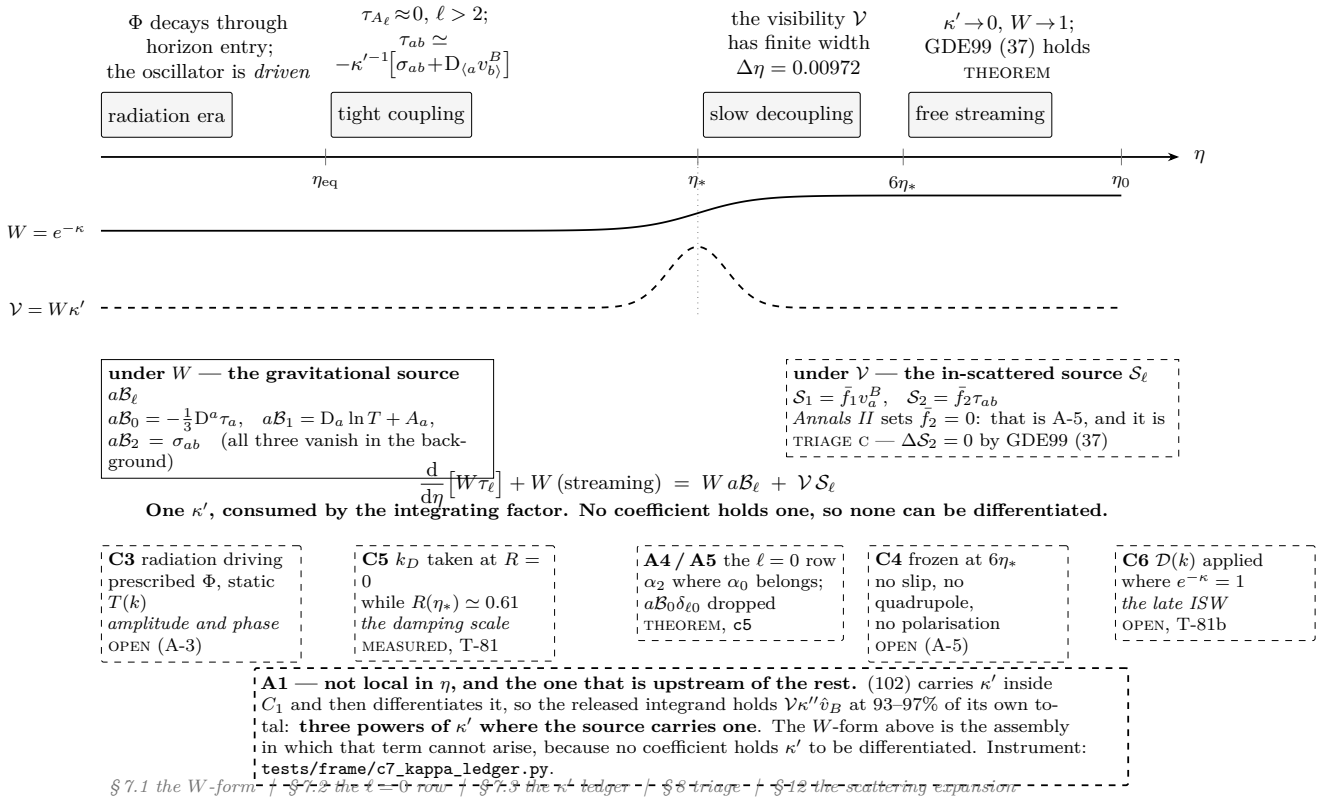


Figure 1: The photon's history and where each open question acts.†

† *The $\ell = 0$ row of* (89) *is corrected here; see findings* **T-104**, **T-105** *in* `provenance/BUNDLE-DEFECTS-v1.md`.

7.1 The W-form

The derivation, in three lines. The covariant transport equation at rank ℓ is

$$\tau'_\ell \;+\; (\text{streaming couplings}) \;+\; \kappa'\tau_\ell \;=\; a\mathcal{B}_\ell \;+\; \kappa'\mathcal{S}_\ell, \tag{1}$$

with $a\mathcal{B}_\ell$ the gravitational source of §*The generic source derivatives*' rank decomposition and $\mathcal{S}_\ell$ the in-scattered Thomson source. Both κ' on display are the same κ': the out-scattering on the left and the in-scattering on the right are two halves of one collision term.

Multiply through by $W = e^{-\kappa}$. With κ accumulated backwards from η_0, so that $\mathrm{d}\kappa/\mathrm{d}\eta = -\kappa'$ and $W' = +\kappa' W = \mathcal{V}$ — the sign this bundle's convention forces, stated at §8 — the first and third terms of (1) combine exactly:

$$W\tau'_\ell + W\kappa'\tau_\ell \;=\; W\tau'_\ell + W'\tau_\ell \;=\; \frac{\mathrm{d}}{\mathrm{d}\eta}[W\tau_\ell]. \tag{2}$$

Hence

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}\eta}[W\tau_\ell] \;+\; W\,(\text{streaming couplings}) \;=\; W\,a\mathcal{B}_\ell \;+\; \mathcal{V}\,\mathcal{S}_\ell} \tag{3}$$

and integrating from η_* to η_0 gives the solution with $\big[W\tau_\ell\big]_{\eta_*}^{\eta_0} = \tau_\ell(\eta_0) - W(\eta_*)\tau_\ell(\eta_*)$ as its only boundary term, since $W(\eta_0) = 1$.

What (3) **says, and it is four things.**

1. **The gravitational source travels under W and the in-scattered source under $\mathcal{V}$.** Neither weight is assigned; both fall out of one multiplication. $\mathcal{V}$ appears because κ' was already beside $\mathcal{S}_\ell$ in the collision term, and $W\kappa' = \mathcal{V}$ is an identity.

2. **κ' is consumed once**, by the integrating factor. It appears nowhere else. *No coefficient holds a κ', so no differentiation can produce a κ'^2.*

3. **There is no integration by parts.** (102)'s bracket $C_0 - C_1' + C_2''$ exists because the by-parts was taken onto the coefficients; (3) has no coefficients to take it onto. The three bulk slots and the boundary pair that §8.4 tries to reassign are artefacts of a step (3) does not take.

4. **The boundary term is one object, and it is exponentially suppressed at η_*.** $[W\tau_\ell]$, not $[C_2' - C_1]$ plus kC_2, and it carries W rather than a diffusion factor. **D18** measures what the released boundary carries instead: with `boundary_weight_w False`, $C_1(\eta_*)$ enters under the diffusion factor alone — no W, no $\mathcal{V}$ — so its bare $\kappa'(\eta_*)$, 3.82×10^2 on Einstein–de Sitter and 2.64×10^2 on `PLANCK_MODEL`, stands under no exponential suppression at all. (3) has no counterpart: $\mathcal{V}\mathcal{S}_1$ is κ'-free.

This supersedes §8.4's item 2, and the wording that item carries is what T-88 implemented. That item reads *"they take different weights: W for the gravitational half, $g = \mathcal{V}$ for the scattering half"*. As a description of the outcome of (3) it is correct. As an instruction to a solver it is not, and it was read as one: multiplying $C_1^{\rm thom} = -\kappa'\hat{v}_B$ by $\mathcal{V}$ applies a second κ', which §8.8 measures at 320 to 682. **The scattering half arrives at $\mathcal{V}$ weight carried on the undifferentiated $\hat{v}_B$; it is not multiplied by $\mathcal{V}$.** Finding **T-103**; the phrasing is the Paper 1 stream's and the correction is recorded against it rather than against the implementation that followed it.

What $\mathcal{S}_2$ is, and why it cannot be a frame defect. *Annals II*'s collision term is $C[\tau] \simeq \sigma_{\rm T} n_E(\bar{f}_1 e^a v_a^B + \bar{f}_2 e^a e^b \tau_{ab} - \tau)$ with $\bar{f}_1 = 1$ and $\bar{f}_2 = 0$, so $\mathcal{S}_2 = 0$ in this reconstruction. That is approximation **A-5** stated at the collision term rather than at the oscillator. **It cannot produce a frame disagreement at any ℓ**: $\mathcal{S}_2 \propto \tau_{ab}$, and GDE99 (37) makes τ_{A_ℓ} invariant for $\ell \geq 2$, so $\Delta\mathcal{S}_2 = 0$ identically and dropping the term changes both columns by the same amount. It is a **§8 Triage C** site — it can ruin the spectrum and it cannot be seen by a frame instrument — and listing it as a candidate for the frame gap confuses the two questions the triage tables exist to separate.

7.2 The $\ell = 0$ matching row

The rows of (89) are Taylor coefficients of the kernel at the origin, and two of the three use their own rank's normalisation. With $h_\ell = \alpha_\ell j_\ell$ and $\alpha_\ell = (2\ell+1)/\beta_\ell$ — this bundle's own `functions/harmonics/weights.py`, where $\beta_\ell = \ell!/(2\ell-1)!!$ gives $\beta_0 = \beta_1 = 1$ and $\beta_2 = \frac{2}{3}$ — each row of (89) pairs C_n with $h_\ell^{(n)}(0)$:

row	used	its own	the entry
$\ell = 2$	$\alpha_2 = \frac{15}{2}$	α_2	$h_2''(0) = \frac{15}{2} \cdot \frac{2}{15} = 1$
$\ell = 1$	$\alpha_1 = 3$	α_1	$h_1'(0) = 3 \cdot \frac{1}{3} = 1$
$\ell = 0$	$\boldsymbol{\alpha_2 = \frac{15}{2}}$	$\boldsymbol{\alpha_0 = 1}$	$-\frac{5}{2}$ printed; $h_0''(0) = -\frac{1}{3}$

(89) writes its $\delta_{\ell 0}$ entry as $5\beta_2^{-1} \cdot \frac{1}{3}$, which is α_2 times $j_0''(0)$ up to the sign **B-11** corrects. The rank-two normalisation is right where it belongs and is carried into a row that is not rank two. **The two rows that are exactly unity are the control**: a normalisation error cannot produce 1 twice and then $-\frac{5}{2}$ once. Finding **T-104**; the discrepancy is $\alpha_2/\alpha_0 = \frac{15}{2}$.

This route is independent of (89), which is what makes it a check rather than a restatement. β_ℓ is fixed by the PSTF normalisation $O^{A_\ell}O_{A_\ell}$ and $j_0''(0) = -\frac{1}{3}$ by the ascending series; neither comes from (89). (89) is the *claim* about what the rows hold and the kernel is the *object*. Two of the three rows reproduce (89)'s entry exactly from the object, which validates the route; the third does not.

The $a\mathcal{B}_0\delta_{\ell 0}$ row is not optional, and restoring it is the other half of the same correction. The paragraph *"Rank 0: not needed"* of §*The generic source derivatives* is **withdrawn**. With α_0 on the left and the source row on the right,

$$C_0 + k^2 C_2\, h_0''(0) = -a\mathcal{B}_0, \qquad h_0''(0) = -\tfrac{1}{3}, \tag{4}$$

and with $C_2 = -k^{-2}a\mathcal{B}_2$ this solves to

$$\boxed{C_0 \;=\; -\tfrac{1}{3}\,a\mathcal{B}_2 \;-\; a\mathcal{B}_0} \tag{5}$$

in place of (90)'s $C_0 = -\frac{5}{2}a\mathcal{B}_2$. Both corrections are needed and neither alone suffices: the rank-2 slot residual γ is $-\frac{5}{2}$ as printed, $+\frac{1}{3}$ with α_0 alone, and 0 only with both. Instrument `c5_monopole.py`, under `tests/frame/`, which also carries a negative control on each alternative.

The corrected row makes C_0 frame-invariant, which answers Appendix 13's open question 3. That appendix finds $\Delta C_0 = +\frac{5}{2}kv$ with no partner anywhere in (102), and measures the consequence at three to four orders of magnitude. Under (5) the shift is instead

$$\Delta C_0 = -\tfrac{1}{3}\,\Delta(a\mathcal{B}_2) - \Delta(a\mathcal{B}_0) = -\tfrac{1}{3}(-kv) - \tfrac{1}{3}kv = 0, \tag{6}$$

using the appendix's own frame table for $a\mathcal{B}_2 = -kv$ and, for the monopole, $\mathcal{B}_0 = -\frac{1}{3}\mathrm{D}^a\tau_a$ with $\tilde{\tau}_a \approx \tau_a - v_a$, giving $a\,\Delta\mathcal{B}_0 = +\frac{1}{3}kv$. ΔC_0 **has no partner because it is zero.** Finding **T-105**. Appendix 13's questions 3 and 4 close with it; its questions 1 and 2 are settled by **T-66** and by §7.1 respectively.

One thing (5) **does not disturb.** Appendix 13 excludes a comoving/physical or amplitude/potential mismatch by measuring $C_0/(k^2C_2)$ as a background-independent and k-independent rational. That argument holds under (5) unchanged — the ratio is still a pure rational, now $\frac{1}{3}$ rather than $\frac{5}{2}$ — because what it rests on is that C_0 and C_2 divide one $a\mathcal{B}_2$. **The measurement was right and what it established was internal consistency, which is not correctness.** It is the same reading the bundle gave T-82 and T-85, applied to itself.

7.3 The κ' ledger

The rule. *κ' may live in W, in $\mathcal{V}$, or in a derivative of either — never in a coefficient.* A coefficient carrying a bare κ' manufactures κ'^2 the moment it is differentiated. At η_* in this bundle's units **D18** measures $\kappa' = 3.82 \times 10^2$ on Einstein–de Sitter and 2.64×10^2 on `PLANCK_MODEL`, so a spurious power of it is two to three orders.

Why it is a rule and not a preference. It separates two κ'^2 that are indistinguishable in the output. The *legitimate* one is inside $\dot{\mathcal{V}} = (\kappa'' + \kappa'^2)W$ and comes from differentiating the **weight**; it is required, and (3) produces it whenever a streaming term is integrated. The *illegitimate* one comes from differentiating a **coefficient**, and nothing requires it. Only the provenance distinguishes them, so the rule is stated on the provenance.

(102) as printed violates it, and this supplement prints both halves of that without reconciling them. §*The generic source derivatives* gives $-C_1 \simeq k^{-1}(a\mathcal{B}_1 + \kappa' v_B)$ — a coefficient holding a bare κ' — and (102) then differentiates it. §8 measures the result: $\mathcal{V}\kappa''\hat{v}_B$ is 93–97% of the released integrand, three powers of κ' where the physical Doppler source $g\hat{v}_B = W\kappa'\hat{v}_B$ carries one. **The definition and the rule are both correct and they contradict each other**; (3) is the assembly in which the contradiction does not arise, because C_1 does not exist in it.

It is enforced before implementation, not after. `tests/frame/c7_kappa_ledger.py` grades an assembly structurally — for each term, whether its coefficient holds κ' and at what power — rather than measuring an output. It reports the W-form clean, and (102) as printed and T-88's variant as carrying κ' in a coefficient at power two. **Both of the defects of 19–20 October would have been caught by that one test before either was implemented**, which is the argument for a structural check over a numerical one: a numerical check on an assembly that is wrong in this way measures the defect faithfully and reports a number.

7.4 What this section establishes, and what it does not

Established. That (3) follows from (1) by one multiplication; that it assigns both weights without a choice and leaves no κ' in any coefficient; that (89)'s $\ell = 0$ row carries α_2 where α_0 belongs, with the $\ell = 1$ and $\ell = 2$ rows as controls; that (5) is the row solved with both corrections and that it makes $\Delta C_0 = 0$; and that $\mathcal{S}_2$ is a Triage C site and cannot carry a frame gap.

Not established. That (3) closes the frame gap. It is the only assembly in the register that satisfies the ledger, and no setting of `integral_solution`'s seven controls reaches it — each redistributes weights while leaving κ' inside C_1 — but that is an argument from exhaustion of what has been tried, not a measurement.

The acceptance condition, stated in advance so the repair can fail visibly. (i) $A_N = A_C$ to the quadrature floor **T-85** measures, neither better nor worse; (ii) the Newtonian column moves by no more than that floor; (iii) `c7_kappa_ledger.py` reports the assembly clean *by construction*, no coefficient holding κ', rather than clean by measurement; (iv) `c5_monopole.py` reports $\gamma = 0$ on the live coefficients and not only on the instrument's own; and (v) **the spectrum adjudicates, and it adjudicates last**, since two wrong columns can agree.

The order is forced, not preferred — in implementation, not in exposition. §7.2 is implemented before §7.1: the $\ell = 0$ row sits inside the bracket the W-form reweights, so a weight correction applied over a C_0 that is wrong by $\frac{15}{2}$ measures the wrong thing. **T-96**'s second run is that measurement — a W-form instrument run against coefficients still carrying the printed row, returning $O(1)$ — and it is reported there as not conclusive, correctly.

8 The frame discrepancy: localisation and triage

This section records the state of the one open defect that the $v1.3.0$ line exists to close. It is written to be auditable rather than persuasive: every exclusion below carries the instrument that licenses it and a grade, because an exclusion without an instrument is an opinion and this project has paid for several. The measurements are recorded in `provenance/BUNDLE-DEFECTS-v1.md`, which is authoritative; numbers appear here only where the argument cannot be followed without them.

8.1 The statement being tested

For $\ell \geq 2$ the covariant temperature multipoles are invariant under a change of the fundamental congruence at linear order. Writing $\tilde{u}^a = u^a + v^a$ with $v^a u_a = 0$ and v^a first order, GDE99 (37) gives

$$\tilde{\tau}_{A_\ell} \;=\; \tau_{A_\ell} \;-\; v_{A_\ell}, \qquad v_{A_\ell} \equiv 0 \quad \text{for} \quad \ell \geq 2, \tag{7}$$

so only the monopole and the dipole move, and every $\ell \geq 2$ multipole is a frame invariant. The bundle computes the same C_ℓ twice — once in the Newtonian frame, $\tilde{\sigma}_{ab} \simeq 0$, and once in the CDM frame — and the two must agree. **They do not**: the disagreement is total, median 9.99×10^{-1} and worst 1.00 over the released multipole set.

Equation (7) is a theorem at linear order. It admits no tolerance and no background dependence, so a measured disagreement is a defect in the implemented specialisation and cannot be read as a limitation of the formalism. That is the premise of everything below.

8.2 Triage A — sites the discrepancy cannot come from

Each row names the instrument and a grade. THEOREM is exact at linear order; MACHINE ZERO is measured at 10^{-13} or better and is structurally zero; MEASURED is a number with a stated validity condition; ASSERTED is read or reasoned and is usable but not load-bearing.

site	instrument	grade
Free streaming and transport	GDE99 (37)	THEOREM
The observer endpoint: a pure dipole	GDE99 (37); Paper 1 §§5.3, 6.2	THEOREM
The Bessel projection	T-71, synthetic control	MEASURED
The C_1–C_2 boost relation	T-77, 3.3×10^{-16}	MACHINE ZERO
C_0's boost behaviour with $\mathcal{B}_0$ restored	T-77, `c5_monopole.py`	MACHINE ZERO
The by-parts identity itself	`c3_byparts.py` and two derivations	MACHINE ZERO
The oscillator at leading order	register; two external reads	MEASURED
The source's frame consistency	**T-82**	RESOLVED
Boundary-against-bulk bookkeeping	**T-85**	RESOLVED, EXCLUDED
The additive $\kappa'\tau_0$ route	**T-79**	RESOLVED, CLOSED

Nothing in this table can be moved by a measurement. A result that appears to contradict one of these rows is wrong, or the comparison is void — most often the latter, through $\ell \gtrsim k\Delta\eta$, where the spherical Bessel function is exponentially small and a ratio of two vanishing numbers is still a number.

8.3 The two measurements that closed the upstream end

The source agrees between the frames (T-82). CAMB carries the rank-1 scattering source as a single object $g\,(v_b + \sigma)$. Under a boost $v_b \to v_b - v$ while the shear term shifts oppositely, so the sum is invariant; a code that never separates them has no frame gap by construction. This bundle does separate them, so the two columns must produce the same sum. In this bundle's slots the combination is $\hat{v}_B + C_2$, with $\hat{v}_B = \tau_1/k$ the reduced dipole and $C_2 = v/k$, since $a\mathcal{B}_2 = a\sigma = -kv$ and $C_2 = -a\mathcal{B}_2/k^2$. Differencing the two columns, on identical histories and before any projection — no transfer function, no j_ℓ, no weight, no damping:

$$\left|\Delta(\hat{v}_B + C_2)\right| \;\in\; \left[\,9.3 \times 10^{-16},\; 1.8 \times 10^{-13}\,\right], \qquad k \in \{1, 10, 60, 300\}, \tag{8}$$

on both the Einstein–de Sitter and the Planck backgrounds. **The source is frame-consistent to fourteen significant figures.** Set against a frame gap of 100% at the spectrum, the pair says what neither says alone: everything between the source and the spectrum is the projection.

The by-parts identity holds in both frames (T-85). With $W = e^{-\kappa}$ and $\mathcal{V} = W'$ — the sign this bundle's κ convention forces, κ being accumulated backwards from η_0 — the transport source obeys, exactly,

$$\int \mathcal{V}\, M\, j_\ell \,\mathrm{d}\eta \;=\; \Big[\, W M j_\ell \,\Big] \;-\; \int W M' j_\ell \,\mathrm{d}\eta \;+\; k \int W M\, j'_\ell \,\mathrm{d}\eta, \tag{9}$$

where $x = k(\eta_0 - \eta)$, so $\mathrm{d}j_\ell/\mathrm{d}\eta = -k\, j'_\ell(x)$ and the last sign is $+k$. Evaluating the left side (call it A) and the right side (B) on identical histories and identical endpoints, with $M = -C_1$, in each frame gives four numbers. Both identities hold: $|A - B|/|A|$ sits at 10^{-3}–10^{-2} and *falls under grid refinement* — $2.83 \times 10^{-3} \to 2.20 \times 10^{-3} \to 2.04 \times 10^{-3}$ at $(k, \ell) = (10, 2)$, and $1.105 \to 0.279 \to 0.0727$ at $(300, 2)$, about $4\times$ per doubling, which is second order and is what the trapezoidal rule gives on an oscillatory integrand.

Since A and B are the same integral by an exact identity, a difference has only two available explanations, and refinement removes one of them. **Boundary-against-bulk bookkeeping is excluded.** But $|A_N - A_C|$ is 1.00 to 1.78 of scale in every valid cell, with the sign reversed between columns: the coefficient definitions carry the whole of the gap.

One apparent corroboration, withdrawn. At low k the ratio A_C/A_N is independent of ℓ to four figures. That carries no information about the frames. $\mathcal{V}$ is a narrow spike, so $\int \mathcal{V} f j_\ell \simeq f(\eta_*)\, j_\ell(x_*) \int \mathcal{V}$ for *any* f, and the ratio of two $\mathcal{V}$-weighted integrals is $f_C(\eta_*)/f_N(\eta_*)$ — free of ℓ by construction. Predicted against measured, the two agree to 0.9–4.2% for $k \le 60$ and fail at $k = 300$, where the mode oscillates inside the visibility width and the ℓ-freedom correspondingly disappears. The check cost one run and removed a finding rather than adding one.

8.4 Triage B — the one frame-sensitive site that remains

Five candidates were open on 18 October. Four are now subsumed or excluded: $\kappa' v_B$ inside C_1 folds into the slot question; the $\ell = 0$ matching row is landed; boundary-against-bulk is excluded by T-85; the dipole boost $-v/k$ and the supplied C'_1 are both subsumed by T-82, which shows the source they would have corrupted is clean. **One site remains: the slot and weight assignment of C_1.**

Identity (9) has three terms and every one of them carries the weight W. The bundle has two, and neither carries W:

	the identity requires	`integral_solution.py` **has**
boundary	$\big[WMj_\ell\big]$, weight $W = e^{-\kappa}$	$(C_2'^* - C_1^*)\, \mathcal{D}(\eta_0, k)\, j_\ell(x_*)$ — weight $\mathcal{D}$, a *diffusion* factor, carrying neither W nor $\mathcal{V}$
j_ℓ bulk	$-\int WM' j_\ell$, weight W, *minus*	$\int \mathcal{V}\mathcal{D}\,\{C_0 - C'_1 + C''_2\}\, j_\ell$ — weight $\mathcal{V}\mathcal{D}$, *plus*
j'_ℓ bulk	$+k \int WMj'_\ell$, M *undifferentiated*	**absent. The bulk has no j'_ℓ slot.**

The structural half is the third row. Everything was integrated by parts *onto the coefficients* — $\{C_0 - C'_1 + C''_2\}$ is (102)'s printed bracket, and it is the W-weighted by-parts form of $\int W\big[C_0 K + C_1 K' + C_2 K''\big]$ — and that decomposition is then evaluated at g weight, where it is not available. The boundary meanwhile is *Annals II*'s own `visibe` collapse: under $\int \mathcal{V} = 1$ the $-\mathcal{V}C_1$ term becomes $C_1(\eta_*)$. **So the bundle carries one term from each of two incompatible decompositions**, and then applies a third weight, $\mathcal{D}$, to one of them.

What the repair is, stated so it can be refuted. Not a split of the slot. A split, *plus* a weight change on each half, *plus* a slot the bulk does not currently have:

1. C_1's two contents separate — M's by-parts residue, and $(v_b + \sigma)$.
2. They *arrive under* different weights: W for the gravitational half, $\mathcal{V}$ for the scattering half. **That is an outcome, not an operation, and reading it as one is finding T-103** — the scattering half arrives at $\mathcal{V}$ weight carried on the *undifferentiated* $\hat{v}_B$; it is not *multiplied* by $\mathcal{V}$. Multiplying $C_1^{\mathrm{thom}} = -\kappa' \hat{v}_B$ by $\mathcal{V}$ applies a second κ', which is the 320–682 of §8.8. See §7.1, where the weights fall out of the integrating factor and nothing is multiplied by hand.

3. The W-weighted half needs *both* bulk slots, j_ℓ and j'_ℓ, because by-parting onto the coefficients is what removed the second, and that step is legal only at W.

The numerics for the missing slot already exist in the file: $j'_\ell = [\ell j_{\ell-1} - (\ell+1) j_{\ell+1}]/(2\ell+1)$ is built inside the rank-2 *boundary* term, which has never been evaluated because C_2 vanishes in every Newtonian run.

The acceptance condition is stated in advance. The split must leave A_N and A_C equal, and must not move the Newtonian column by more than the quadrature floor T-85 measured. Frame agreement alone is *not* sufficient — two wrong columns can agree — so the spectrum adjudicates, and it adjudicates last.

8.5 Triage C — frame-blind sites

A term X with $\Delta X = 0$ passes every frame test for free, so no frame instrument can see it. These cannot cause the discrepancy and can ruin the spectrum; they are listed so that the two questions are not confused.

	site, and what it bears on	**status**
C1	the coefficient of any boost-invariant term; low-k amplitude	`c5` blind by construction
C2	`LCDM_MODEL` without Ω_r; peak positions	confirmed, 37%, T-83
C3	radiation-era driving, static $T(k)$; amplitude and phase	open (A-3)
C4	the tight-coupling freeze at $6\eta_*$; peak contrast, temperature–Doppler balance	open (A-5)
C5	k_D computed at $R = 0$; one $\mathcal{D}(k)$ across histories; higher peaks and the tail	T-81a/c/d, v1.4.0
C6	$\mathcal{D}(k)$ applied to the integrand *at all*; the late ISW	T-81b — first named mechanism

C6 and the slot question are the same structural change seen twice. Diffusion damping is a scattering effect; gravitational redshift accumulated after decoupling is not damped by photon diffusion. Scattering damping belongs to the scattering half of the source and not to the gravitational half — so the split that fixes the weights fixes the damping in the same stroke. One change, three symptoms: the frame gap, the missing late ISW, and the $\mathcal{D}$-on-the-boundary mismatch.

8.6 Triage by conformal time

The same question asked along the photon's history. It localises cleanly, and the over-damping sits in two windows — neither of which is an integration limit, since T-85 settled the bookkeeping.

window	**what the code does**	**what it costs**
$\eta < \eta_{\rm eq}$	prescribed matter-growth Φ times a static $T(k)$, where the potentials should decay through horizon entry and drive the oscillator	peak amplitude and phase (C3)
$\eta_{\rm eq}$	no Ω_r in `LCDM_MODEL`	peak positions, 37% in ℓ_A (C2)
$\eta_{\rm eq} \to \eta_*$	oscillator sound speed correct; k_D evaluated at $R = 0$ while the oscillator runs at $R(\eta_*) \simeq 0.61$	the damping scale (C5)
$\eta_* \pm \frac{1}{2}\Delta\eta_{\rm vis}$	instantaneous boundary, envelope substituted for a visibility of finite width $\Delta\eta = 0.00972$	the tail from $\ell \simeq 785$
$\eta_* \to 6\eta_*$	tight-coupled oscillator continued and then frozen; no slip, no growing quadrupole, no polarisation	peak contrast, the temperature–Doppler balance (C4, A-5)
$6\eta_* \to \eta_0$	the integrand carries $\mathcal{D}(k)$ where $e^{-\kappa} = 1$ and no scattering occurs	the late ISW (C6)
η_0	reception: Doppler and aberration	nothing — THEOREM

Three consequences are worth drawing out.

The integration range is clean. T-85 settled the bookkeeping, so how far the problem has been integrated is no longer a candidate.

"Conformal time" and "missing radiation" are one question. η is not a free choice: it is fixed by the background. A radiation-free background gives a different $\eta(z)$, hence a different $k\Delta\eta$ at every point, hence different Bessel arguments and different peak positions. The 37% in ℓ_A *is* the conformal-time defect, already measured.

Over-damping is two substitutions, not a limits error, and both sit after η_*: an envelope standing in for a visibility of finite width, and $\mathcal{D}(k)$ applied to a source generated where no scattering occurs. A discriminator is available and was not used until now: Silk damping bites at $\ell \simeq 1807$, the visibility full-width scale is $\ell \simeq 785$, and the computed spectrum departs at $\ell \simeq 700$. **The departure fingerprints the width scale, not Silk** — so k_D at $R = 0$ is a real defect and is *not* what sets where the tail goes wrong.

8.7 What would change this account

Two outcomes would move it, and both are cheap.

If the slot read does not match the shape in §8.4, the discrepancy returns to having no named site, which would be the first time since 18 October. *It matched*; the table in §8.4 is that read.

If removing $\mathcal{D}(k)$ from the integrand alone recovers a late ISW, C6 is confirmed and C3–C5 reduce to amplitude-only concerns.

8.8 The repair, attempted five times, and withdrawn

Recorded here because five failures carry more information than one success would, and because a reader of this document should not have to reconstruct them from the register.

T-86's reading was implemented as four independent controls, every one defaulting to `False`. The first combination closed the frame gap by four orders at $\ell \geq 50$ for $k \lesssim 60$ — and **is withdrawn**. It applied $\mathcal{V}\, C_1^{\rm thom}$ where (102) applies $W\, C_1^{\rm thom}$, and since

$$C_1^{\rm thom} = -\kappa'\hat{v}_B, \qquad W\kappa' = \mathcal{V} \quad\Longrightarrow\quad W\, C_1^{\rm thom} = -\mathcal{V}\hat{v}_B \textbf{ identically,}$$

the Thomson term is already at visibility weight in (102) as printed and there was never a weight to change. The control therefore multiplied it by a spurious κ', measured at 320 to 682. The diagnostic spectrum it produced — first peaks moving from 1.3–1.5 ℓ_A to 0.56 ℓ_A — is withdrawn with it and is retained in `diagnostics/` as the record of what the defective variant produces, not as evidence.

What the five failures leave standing is a rule, and it is P1-T's: κ' may live in W, in $\mathcal{V}$, or in a derivative of either — *never in a coefficient.* A coefficient carrying a bare κ' manufactures κ'^2 the moment it is differentiated. This separates the *legitimate* κ'^2 inside $\dot{\mathcal{V}} = (\kappa'' + \kappa'^2)W$, which comes from differentiating the **weight**, from the illegitimate one, which comes from differentiating a **coefficient**; both look identical in the output. It is enforced structurally by `tests/frame/c7_kappa_ledger.py`, and **both of this week's defects would have been caught before implementation by that one test.**

8.9 What one term does to the integrand

Measured on the released code, as shares of the summed absolute Newtonian integrand:

term	EdS k=10	k=60	k=300	PLANCK k=10	k=60
$\mathcal{V}\kappa''\hat{v}_B$	**0.935**	**0.971**	0.616	**0.933**	**0.974**
$\mathcal{V}\kappa'\hat{v}'_B$	0.066	0.024	0.314	0.065	0.015
$\mathcal{V}\frac{k}{3}\tau_1$	0.0002	0.0064	0.082	0.0004	0.013
$\mathcal{V}(\Phi'_A - \Phi'_H)$	0	0	0	0.0018	0.0028

One term is 93–97% **of the integrand**; the acoustic driving is 0.02% and the ISW, where there is one, 0.18%. Against the source it stands for — the physical Doppler $g\hat{v}_B = W\kappa'\hat{v}_B$ — it is 1.81×10^5 (EdS) and 8.99×10^4 (PLANCK), and $\kappa'' \simeq -\kappa'^2$, so it carries **three powers of κ' where the source carries one**.

This is weight-independent. Changing $\mathcal{V} \to W$ divides every term in the bracket by κ' uniformly and changes no ratio, so A-2 cannot reach it. The term exists only because (102) differentiates C_1, which contains κ' — that is, because the integration by parts was taken *onto the coefficients* rather than onto the product.

8.10 What this section does not establish

No measurement here is a comparison against an external code, and none is a claim that the spectrum is right. The chain establishes that the rank-1 source is frame-consistent, that the by-parts identity holds, and that the slot-and-weight assignment is where the remaining disagreement lives. **It does not establish that repairing the assignment closes the gap**, and the acceptance condition in §8.4 is written so that the repair can fail visibly.

9 Comparison and its range

The C_ℓ comparison is Sachs–Wolfe only: (196) descends from (181), in which the acoustic modulation $\cos(kr_s)$ is dropped by taking $r_s^* \to 0$, and §8.3.2 sets the transfer function to unity. Its accuracy is therefore bounded by the term that was dropped. The range is derived, not chosen: $2 \le \ell \le 20$ at 5%, generated by `scripts/derive_ell_range.py`.

The wider comparison cites CMBFAST for its *method* — line-of-sight integration, Seljak & Zaldarriaga, the same reference the antecedent's own bibliography carries — and compares against **CAMB** and **CLASS**, because the Fortran 77 original is unmaintained and effectively unrunnable. Saying so is more honest than claiming a comparison against a code nobody can run. It is reported in three bands, whose failure modes differ — $2 \le \ell < 30$, cosmic-variance dominated and meaningful only binned; $30 \le \ell < 1000$, the acoustic peaks, where a hierarchy or recursion error actually shows; and $1000 \le \ell \le 2000$, the damping tail, where C_ℓ is small and fractional differences blow up on a vanishing signal. The comparison is made against the corrected closed form.[6][7]

9.1 Criterion 5b's tolerance

The tolerance is *measured* rather than chosen; what it is measured against changed once it was run. The original rule took the CAMB–CLASS mutual difference as the floor, on the ground that no agreement can be claimed tighter than two trusted independent codes agree with each other. That floor is 0.370% across the acoustic peaks, and this reconstruction stands at 137% there. It was never going to meet it: (102) is a slow-decoupling solution with no polarisation hierarchy, no neutrino anisotropic stress and a visibility-weighted line of sight, set against two full Boltzmann codes. **A criterion that can only be failed is not a criterion.**

What *is* gateable is the thing the antecedent writes twice and recovers symbolically but never evaluates. Its integral solution appears as (102), carried with a generic u^a and specialised afterwards, and again as the source assembly (177)–(179), written in the Newtonian frame from the start. The paper states the

[6] The reduction of (200) carries a factor of the comoving distance that the displayed (201) does not; substituting $z = k\chi$ gives χ^{2-n}, which is χ^1 at $n = 1$. The companion result (204) descends from a $P(k)$ definition differing by one power of k and is correctly free of that factor, so the asymmetry is not a units convention. A misprint, corrected silently and recorded.

[7] That derivation uses the Bessel identity of §8.3, which requires $[(m/2)!]^2$ in its denominator. As printed it is exact where $\Gamma(m/2+1) = 1$, so the large-scale result (201) is unaffected; the horizon-scale normalisation (204) and the D_ℓ of (205) are. A misprint, corrected silently and recorded.

relation and carries it out on the page; what no one has done is to compute both forms and compare them, which is the only part of the claim arithmetic can settle. Measured here over three backgrounds, nine modes $0.3 \le k \le 600$ and $2 \le \ell \le 600$, the two agree to 9.5×10^{-6} worst and 2.7×10^{-8} median against a gate of 10^{-3}. The comparison is stated at the transfer level at fixed k with the same line-of-sight weight on both sides: at C_ℓ the two branches differ by a factor of eighty in the plateau, and that difference is the weight the line-of-sight integral carries — a property of what (102) prints, recorded as an approximation and measured separately, not a disagreement to be gated away.

The external comparison is still run, still stamped, and reported alongside as context. Its value is not a pass mark but an itemisation: how far a $1 + 3$ slow-decoupling solution stands from a Boltzmann code, in bands, with the dominant term named.

The measurement itself is recorded here rather than only in the code. The external codes are not carried as dependencies of the bundle and are not run by the harness: `scripts/make_reference_spectra.py` regenerates the stamped spectra from CAMB and CLASS, `scripts/compare_criterion_5b.py` reproduces the comparison from those stamps with neither code installed, and the table below is what that script writes. It is a committed artefact of the bundle, recoverable and reproducible, and deliberately outside the bundled testing.

Shape, D_ℓ renormalised at $\ell = 30$. Context, not the gate (S4c).

band	vs CAMB	vs CLASS	CAMB vs CLASS
plateau and late ISW, $2 \le \ell < 30$	97.8%	97.8%	—
acoustic peaks, $30 \le \ell < 1000$	137.0%	137.3%	0.370%
First acoustic peak			
this bundle	CAMB	CLASS	
$\ell = 170$	$\ell = 198$	$\ell = 198$	

Itemised: the part this bundle can attribute. The damping coefficient, 0.1% in the plateau and late ISW and 17.2% in the acoustic peaks, between (dif-damp-correct) at $f_2 = 3/4$ and (dif-damp) at $f_2 = 1$ — both *Annals II*'s own. The remainder is the slow-decoupling order, the visibility weight of (102) (approximations A-1 and A-2), and the acoustic scale, $z_* = 1114.5$ here against CAMB's 1088.5.

What a self-comparison does not establish is stated here rather than left to be inferred, because it is the obvious objection and it is a fair one. Both routes are handed the same background, the same recombination history, the same acoustic solution, the same potentials and the same damping, and a comparison that holds an input fixed cannot check that input. This is not a formality: setting the gate turned up an under-resolved recombination grid that moves the primary source at η_* by 1.2%, shared by both routes and invisible to the comparison between them. It is recorded, raised, and left for the decision it needs rather than changed inside a release.

The external codes of criterion 5b are compared in `provenance/EXTERNAL-CODE-COMPARISON-v1.md`, which records what CAMB and CLASS each model and — more to the point — what the comparison structurally cannot settle. Neither carries the $1 + 3$ covariant formalism, neither distinguishes mode from multipole coefficients, and neither carries the $O(\varepsilon^2 \ell)$ couplings of v1.5.0. So 5b is a check on the *linear* reconstruction and is incapable of validating the nonlinear extension, which is precisely why v1.0.0 must carry no $O(\ell)$ coupling: if one leaked in, the disagreement with CAMB and CLASS would be indistinguishable from a bug. The canonical parameter set switches reionisation, lensing and massive neutrinos **off**, each being physics this bundle does not model.

10 Evidence

Each figure is generated by the script named in its caption and is accompanied by a standalone caption file in `captions/`. This section is itself generated by `scripts/make_figure_pages.py`, so a figure appears

here only once it exists in the tree, and what is outstanding is stated rather than omitted.

11 Release progression

A reproducibility bundle is not only a result; it is a record of how the result was arrived at, including the parts that were wrong on the way. This section is that record, one release at a time. Each entry states what was *achieved*, and then — separately, because they are different things — what was *learned*, meaning a lesson that changed how the next release was built.

11.1 v1.0.0 — the reconstruction, and whether it is faithful

Achieved. The free-streaming chain of Appendix F reproduced against *four* external hierarchies — Hu & Sugiyama Eq. (6), Wilson Eq. (8), Ma & Bertschinger Eqs. (49)/(50), Seljak & Zaldarriaga Eq. (3d) — each read from its own paper and integrated as printed, at residuals of 10^{-13}–10^{-14} against a 10^{-11} threshold, in two normalisations and two phase conventions. The almost-Friedmann–Lemaitre spectrum (186) computed from (176) by the mode route and the covariant route, agreeing exactly. The Sachs–Wolfe limit recovered at 0.26% against a 5% criterion. **This shows that** the reconstruction is faithful to the antecedent's own equations, independently of any Boltzmann code, and that the harmonic phase convention is fixed by the *definition* of Q_{A_ℓ} rather than adopted.

Learned, and it set the pattern for everything after. A coarsened η grid returned a smooth, plausible spectrum that was 575% wrong while the two routes agreed to 10^{-16} — finding **T-3**. **Two agreeing routes agree about what they share.** That sentence is the single most load-bearing lesson in this bundle; it produced `required_n_eta`, it is why criterion 5b's gate states its shared inputs explicitly, and it is why the grid register exists.

11.2 v1.1.0 — the recombination that makes peaks possible

Achieved. Peebles' effective three-level atom replacing equilibrium Saha, and made the default only after being checked standalone against the standard ionisation history. Freeze-out at 2.15×10^{-4} against a standard $\sim 2 \times 10^{-4}$; the visibility peak moves $z \simeq 1300 \to 1107$ against a standard 1080–1100, and the function *widens by about 40%*. The frame change made numerical rather than symbolic. **This shows that** the width, not the position, is what the spectrum needs: the width sets the Silk damping envelope, which is why the acoustic peaks could not be attempted before this release.

Learned. `SAHA` was kept rather than deleted. A known-worse alternative that *must* change the answer is evidence; deleting it would have left the improvement asserted rather than measured. The same principle appears as the negative control in every criterion that claims one.

11.3 v1.2.0 — the spectrum with structure, and the audit of it

Achieved. The full linear C_ℓ: acoustic peaks, a Silk damping tail, the complete source assembly, and the integral solution (102) carried with a generic u^a and specialised back. Criterion 5b closed at 9.5×10^{-6} against a gate of 10^{-3}. **This shows that** *Annals II*'s two writings of its own integral solution — (102) specialised, and (177)–(179) assembled — are the same calculation when evaluated, which the paper recovers symbolically and which nobody had ever computed.

Learned, three times over, and they are the same lesson in three places. *First*, the tolerance this criterion was written against could only ever be failed: a slow-decoupling solution with no polarisation hierarchy cannot meet two full Boltzmann codes, and a criterion that can only be failed is not a criterion (**S4c**). *Second*, the figure 10^{-5} that had stood in the record for the two routes' agreement was never measured — it came from a fixed grid on which both routes were *un-converged in the same direction*

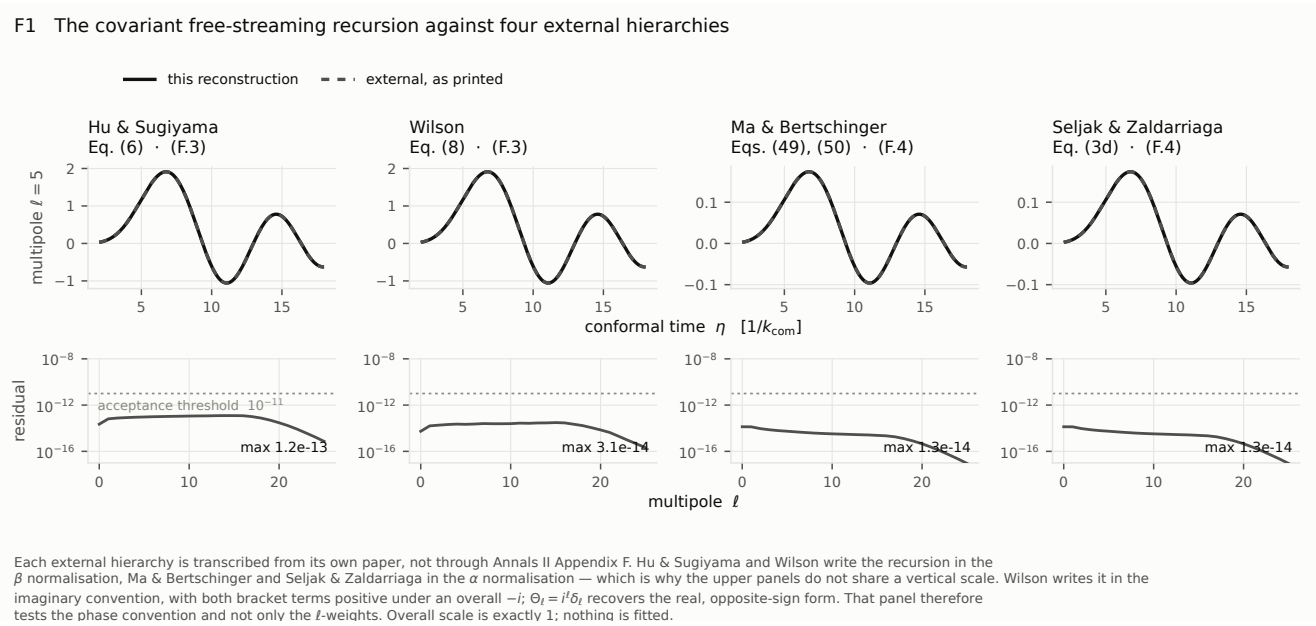


Figure 2: **F1 — The covariant free-streaming recursion against four external hierarchies.** The covariant free-streaming recursion of *Annals II* Appendix F, integrated from a unit monopole, against four external treatments of the same hierarchy. Each is transcribed from its own paper, never through Appendix F. **Upper panels, context:** the reconstruction and the external form at $\ell = 5$; they coincide. The panels do *not* share a vertical scale, because two normalisations are on display. **Lower panels, the claim:** $\max_\eta |\text{external} - \text{reconstruction}|$ against ℓ, with the acceptance threshold marked. Hu & Sugiyama write the recursion in the β normalisation and Ma & Bertschinger and Seljak & Zaldarriaga in the α normalisation, which is why Appendix F prints both (F.3) and (F.4); a basis phase leaking into the couplings would break at least one match. Overall scale is exactly 1.000000000 — nothing is fitted. Four panels from five sources: Wilson & Silk (1981) is the earliest, but at $K = 0$ it is Wilson (1983) Eq. (8) exactly, so it is credited rather than drawn.

Evidence category: cross-check. *Generated by* `scripts/figure_f1_appendix_f.py`. *Standalone caption:* `captions/F1-appendix-f.md`.

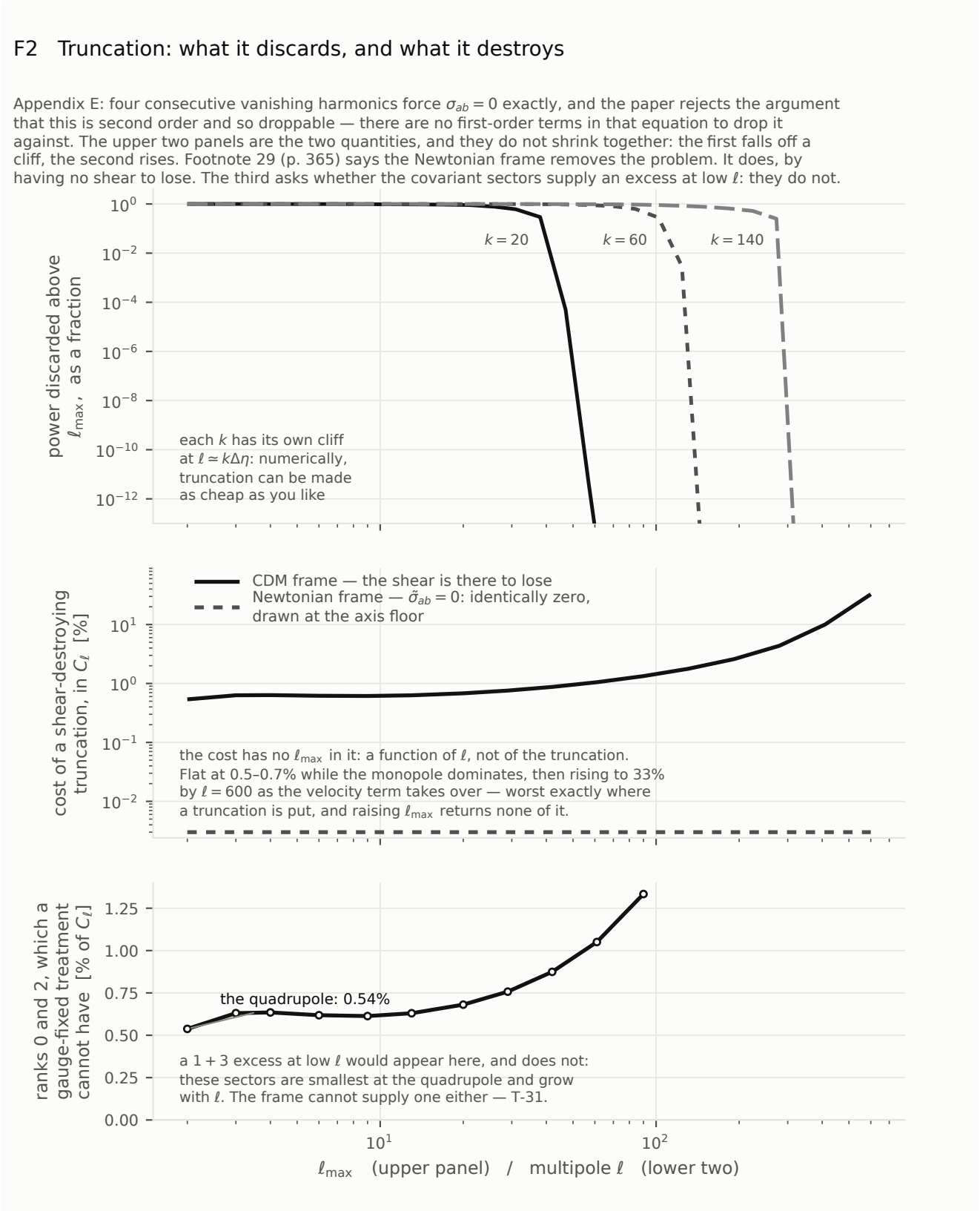


Figure 3: **F2 — What a truncation discards, and what it destroys.** Appendix E of *Annals II* proves that four consecutive vanishing harmonics force $\sigma_{ab} = 0$ exactly. This figure makes that quantitative in both frames, and the two panels answer two different questions that a single "truncation error" would blur.

What a truncation *discards* falls off a per-k cliff at $\ell \simeq k\,\Delta\eta$: above the cliff there is nothing left to keep, and the tail above $\ell = 600$ is identically zero at every probe mode. **What it *destroys*** has no $\ell_{\max}$ in it at all. Setting $\sigma_{ab} \to 0$ costs 0.54–0.68% out to $\ell \simeq 20$, rises monotonically above $\ell = 20$, and reaches 32.6% by $\ell = 600$ — a cost that grows with the multipole rather than being bounded by it, which is why a truncation argument cannot retire the shear.

Footnote 29 (p. 365) is checked and upheld. In the Newtonian frame the cost is *identically* zero

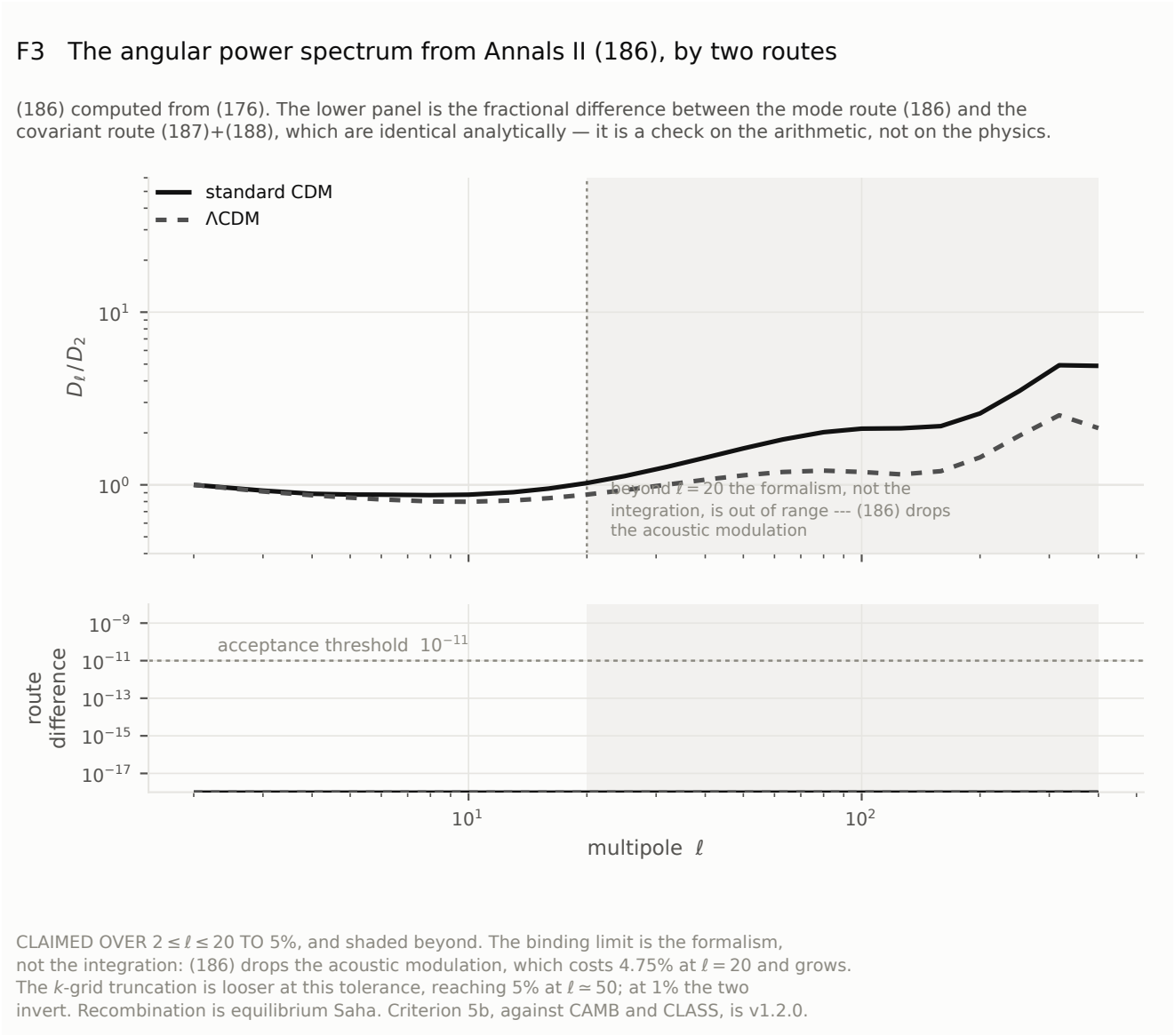


Figure 4: **F3 — The angular power spectrum from Annals II (186), by two routes.** Eq. (186) computed from Eq. (176), for the standard-CDM model of §8.3.2 and for ΛCDM, which *Annals II* does not treat. **Upper panel:** $D_\ell = \ell(\ell+1)C_\ell/2\pi$, each normalised to its own $\ell = 2$ value, so that the comparison is of *shape* and not of an amplitude neither model fixes here. **Lower panel:** the fractional difference between the mode route of (186) and the covariant route of (187) with (188). Those two are identical analytically, so the panel is a check on the arithmetic and not on the physics — and it is worth having, because it caught a factor of four in this bundle's own harmonic weights before any figure was drawn. It sits at 10^{-16}, five orders below the 10^{-11} acceptance threshold.

Evidence category: recovery. *Generated by* `scripts/figure_f3_angular_spectrum.py`. *Standalone caption:* `captions/F3-angular-spectrum.md`.

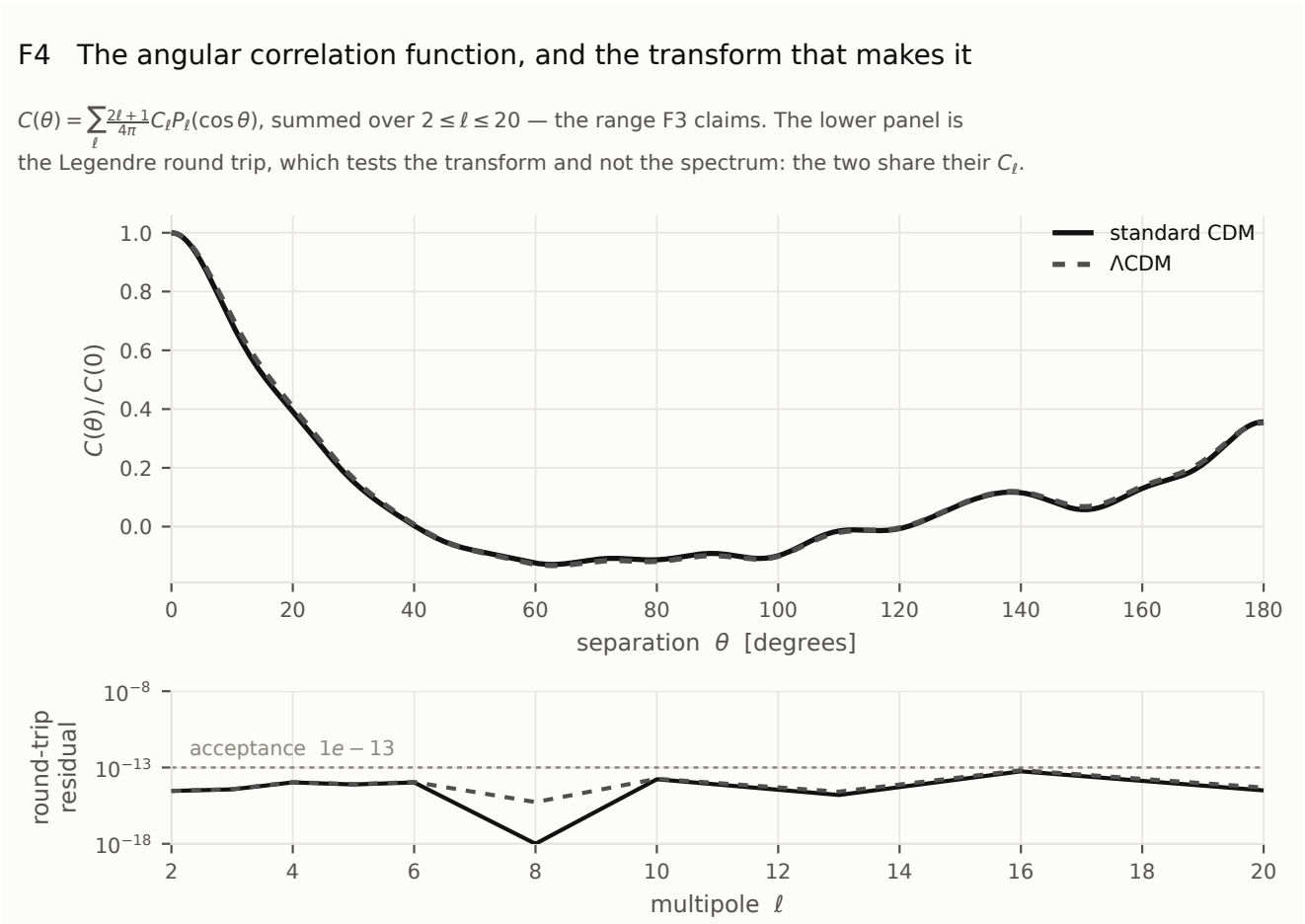


Figure 5: **F4 — The angular correlation function, and the transform that makes it.** $C(\theta) = \sum_\ell \frac{2\ell+1}{4\pi} C_\ell P_\ell(\cos\theta)$, the real-space view of Figure 3: the same content read as a correlation between two directions separated by θ rather than as power at a multipole. **Upper panel:** $C(\theta)/C(0)$ for standard CDM and ΛCDM, summed over $2 \le \ell \le 20$ — the range Figure 3 claims, bounded by (186) dropping the acoustic modulation. Summing further would draw multipoles this release does not stand behind, and a real-space curve hides which ℓ contributed, so the restriction is made in the transform rather than in this caption. **Lower panel:** the Legendre round trip $C_\ell \to C(\theta) \to C_\ell$, at 4×10^{-14} and 7×10^{-14} against an acceptance of 10^{-13}. It tests the transform and *not* the spectrum — the two share their C_ℓ — which is the same caution Figure 3's residual panel carries and for the same reason.

Evidence category: acceptance. *Generated by* `scripts/figure_f4_correlation.py`. *Standalone caption:* `captions/F4-correlation.md`.

F5 The source decomposition in two frames, by rank

The total is common to both frames and the decomposition is not. Split by rank, which is what the match (89) produces and is defined in every frame — not by the primary/Doppler/integrated split of §7.1.1, which exists only after the Newtonian choice has emptied the shear sector. The agreement is measured and it is not exact: the second panel is the gap, and it is drawn rather than described.

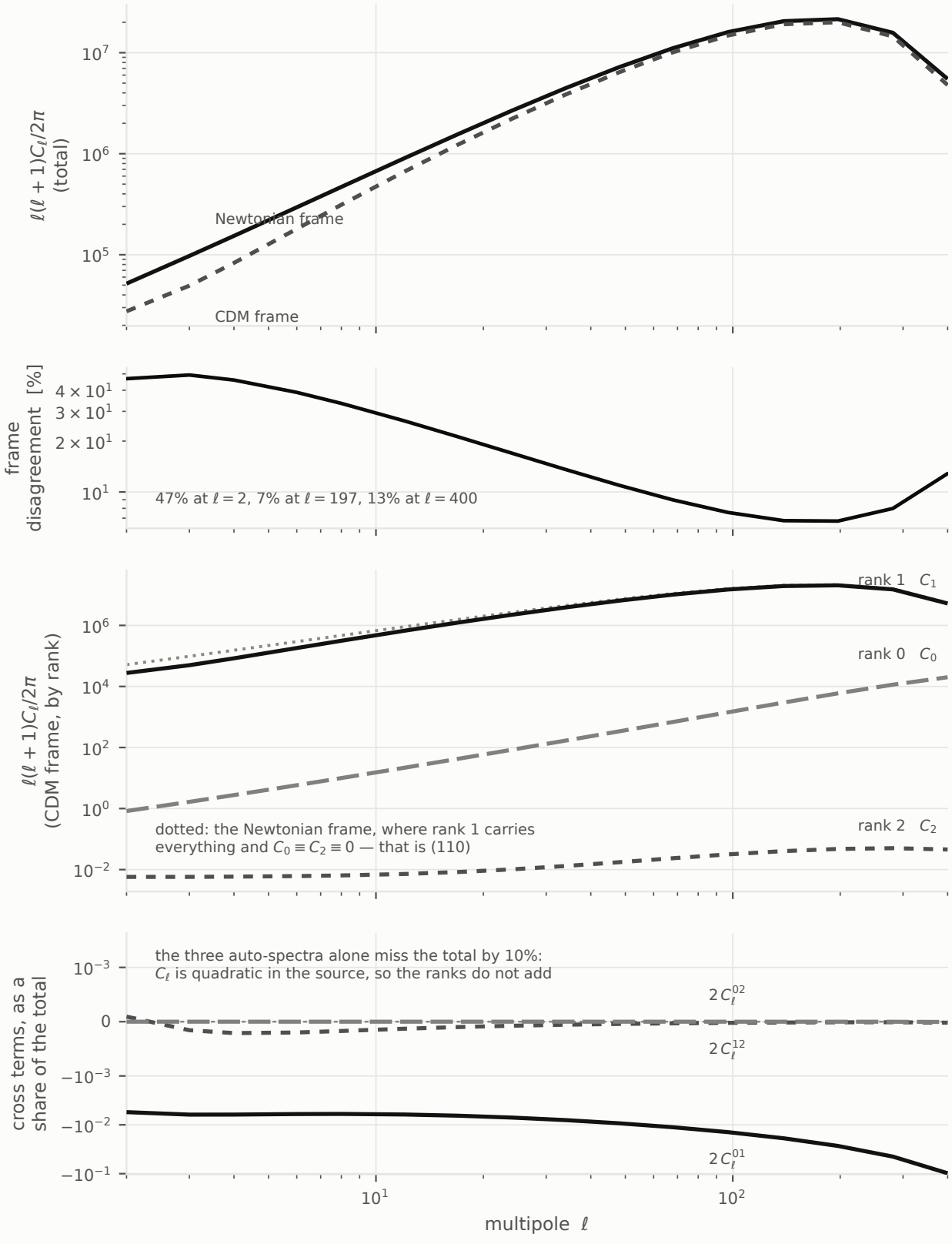


Figure 6: **F5 — The source decomposition, by rank, in two frames.** The integral solution (102) carries three coefficients C_0, C_1, C_2 — one per PSTF rank — and this figure draws them, in the

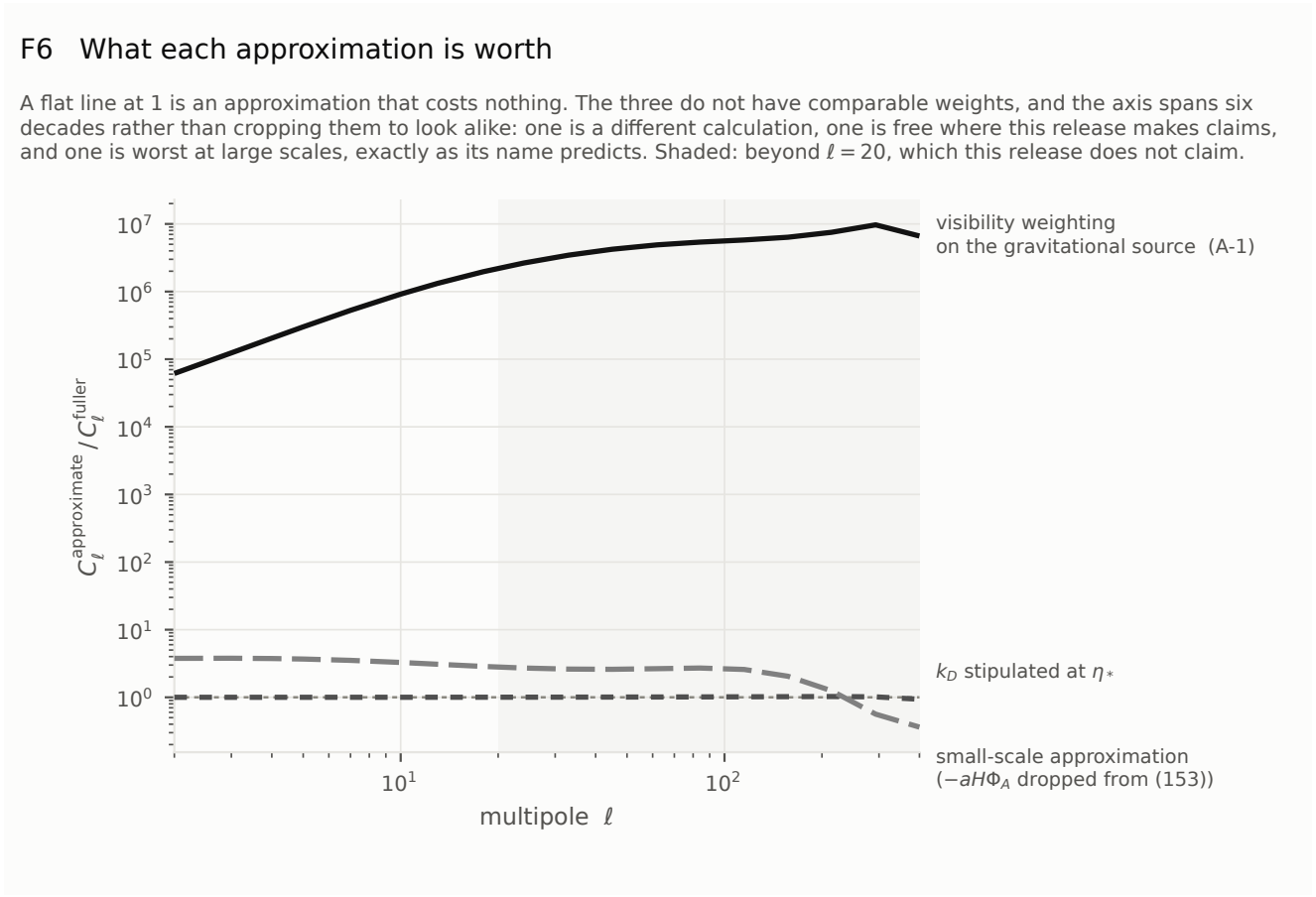


Figure 7: **F6 — What each approximation is worth.** Each series is $C_\ell^{\text{approximate}}/C_\ell^{\text{fuller}}$ for standard CDM, so the panel is scale-free and a flat line at unity is an approximation that costs nothing. *Annals II* makes three narrowing choices which this bundle reproduces rather than corrects, because reproducing a paper means reproducing its approximations (decision S15); they are recorded as **A-n** in `provenance/SOURCE-APPROXIMATIONS-v1.md`, deliberately apart from the defects recorded as **B-n**. This is the one place their cost is a number rather than an adjective.

The three do not have comparable weights, and that is the result, so the axis spans six decades rather than cropping them to look alike. The visibility weighting on the gravitational source (A-1) runs to 10^4–10^6: it is a *different calculation*, not a perturbation of one, which is why `Weighting` has no default. Stipulating k_D at last scattering costs under 0.1% across the claimed range and about 5% by $\ell = 400$ — it **vindicates the release**, which takes the cheap choice where it makes claims. The small-scale approximation, dropping $-aH\Phi_A$ from (153) and so producing the printed oscillator (154), departs from unity most at *large* scales, which is what its name predicts.

A correction kept because it nearly became this figure. The k_D series was first computed with $k_D(\eta)$ along the whole ray. After last scattering $\kappa' \to 0$, so $1/\kappa'$ in (dif-damp) diverges and k_D collapses, damping every integrated source to nothing — a plasma formula evaluated in a universe that no longer has a plasma. That version reported the release's choice as costing a factor of nine at $\ell = 2$; the factor was the artefact. Clamped to $k_D(\min(\eta, \eta_*))$ the comparison is a real one. Shaded: beyond $\ell = 20$, which this release does not claim.

Evidence category: diagnostic. *Generated by* `scripts/figure_f6_approximations.py`. *Standalone caption:* `captions/F6-approximations.md`.

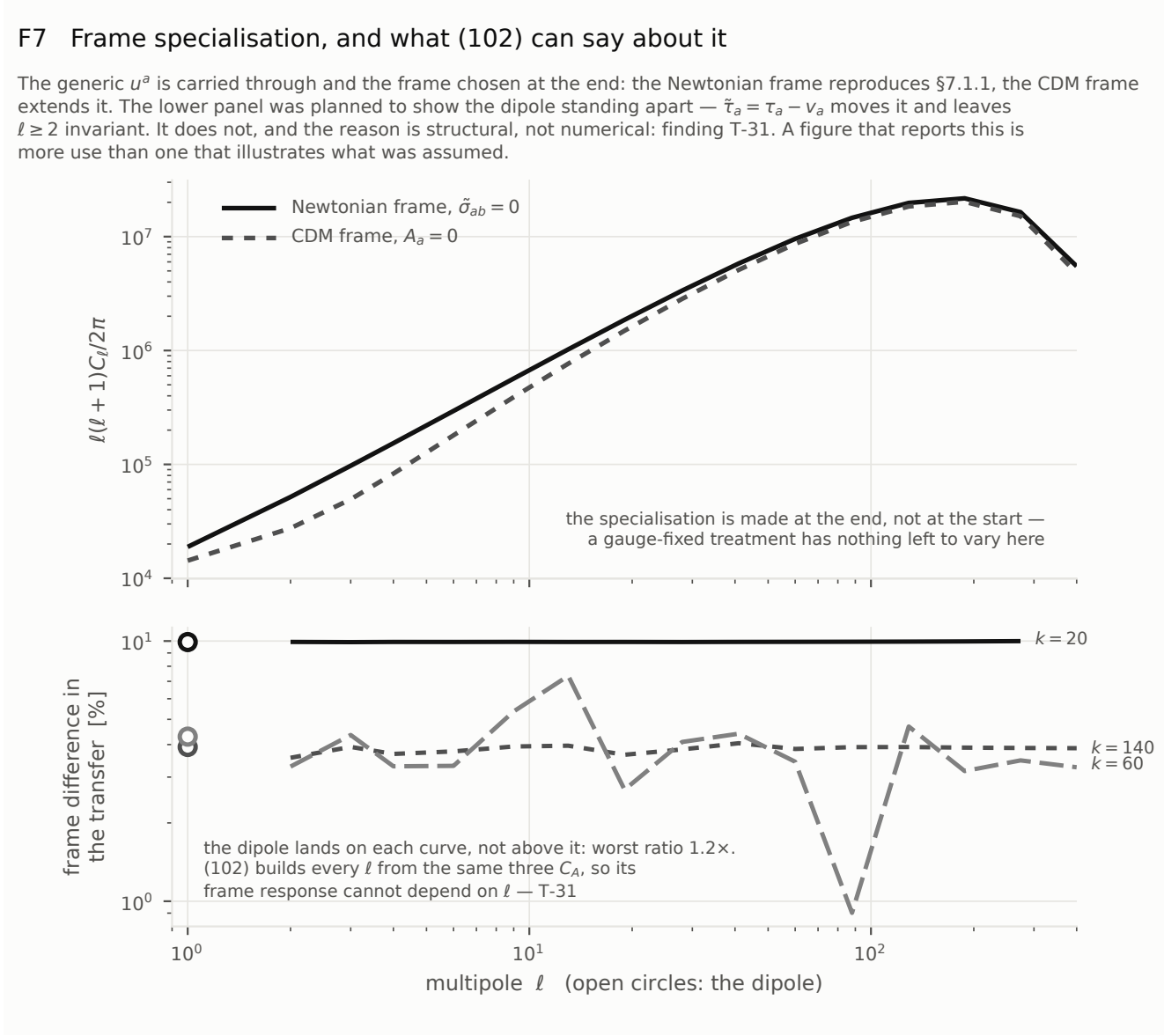


Figure 8: **F7 — Frame specialisation, and a planned claim that was false. This figure reports that its own planned claim is false, and that is why it is here.** It was designed to show the dipole standing an order of magnitude apart from the higher multipoles under a change of frame — the dipole being the one multipole a boost moves at first order. It does not. The dipole lands on each curve instead: worst ratio 1.25× over the probe set, and 1.00× at $k = 20$.

The reason is structural and is finding T-31. Equation (102) builds *every* ℓ from the same three coefficients C_A, so a frame change that moves C_1 moves every multipole built on it in the same proportion. A frame response that depended on ℓ would contradict the equation's own form, and `test_102_cannot_make_its_frame_response_depend_on_the_multipole` asserts that it does not. The spread *over* ℓ does grow with k — 1.01, 1.14, 8.14 at $k = 20$, 60, 140 — but that is the transfer function's k dependence, not the frame's ℓ dependence.

A figure that had been drawn to its plan would have been drawn wrong. The plan is kept in `captions/FIGURE-PLAN-v1.0.0.md` and the outcome is kept here, because a reproduction bundle whose figures only ever confirm their captions is not measuring anything.

Its environment is built for the k grid, not for the probe modes — finding **T-34**, where this figure sized its η grid for $k = 140$ and then integrated to $k = 420$. `check_k` now raises rather than interpolating past what the grid resolves.

Evidence category: diagnostic. *Generated by* `scripts/figure_f7_frames.py`. *Standalone caption:* `captions/F7-frames.md`.

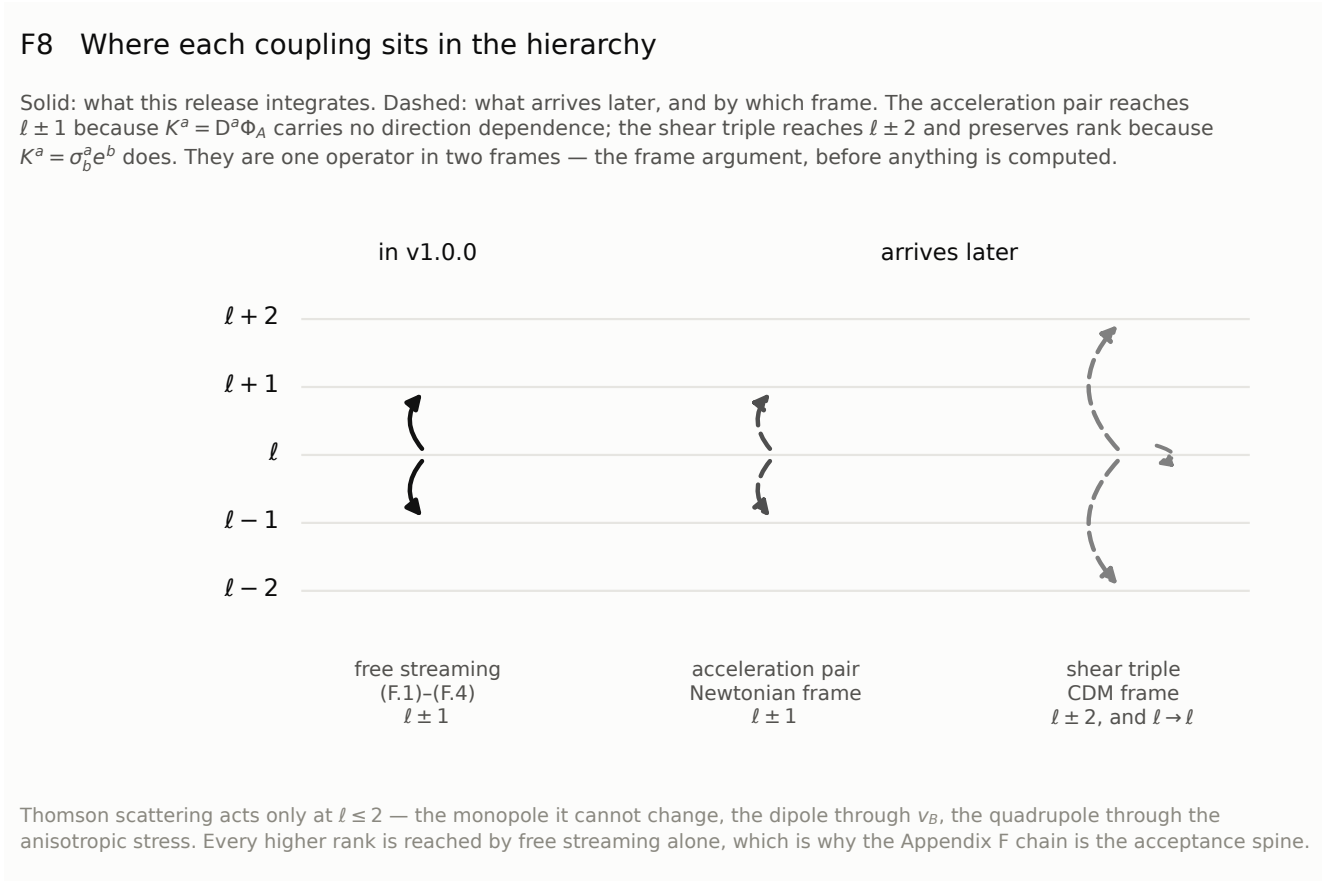


Figure 9: **F8 — Where each coupling sits in the hierarchy.** The one released figure that computes nothing: a statement about the *structure* of the exact divergence hierarchy, read off the equations rather than measured. It is generated by a script like every other figure, so that the manifest covers it — a hand-drawn diagram is the one artefact that can drift without any gate noticing.

Solid: what this release integrates. Free streaming, (F.1)–(F.4), couples rank ℓ to $\ell \pm 1$ and to nothing else, which is the whole of v1.0.0's hierarchy and the reason the Appendix F chain is the acceptance spine. **Dashed:** what arrives later, and by which frame. The acceleration pair reaches $\ell \pm 1$ because $K^a = \mathrm{D}^a\Phi_A$ carries no direction dependence of its own; the shear triple reaches $\ell \pm 2$ *and* preserves rank because $K^a = \sigma^a{}_b e^b$ does. The rank-preserving coupling exists only when K^a depends on direction, and is the single structural difference between the two frames.

Both are the same operator $K^a_\perp \partial/\partial e^a$, and the number of ranks each reaches is fixed by that one property. **That is the whole content of the frame argument, visible before any of it is computed.** Decision S9 is drawn as well as asserted: every coupling in the solid half is a ratio of linear polynomials and tends to a constant, so an $O(\ell)$ coupling wired in early would fail `tests/test_appendix_f_chain.py` at large ℓ.

Evidence category: schematic. *Generated by* `scripts/figure_f8_schematic.py`. *Standalone caption:* `captions/F8-schematic.md`.

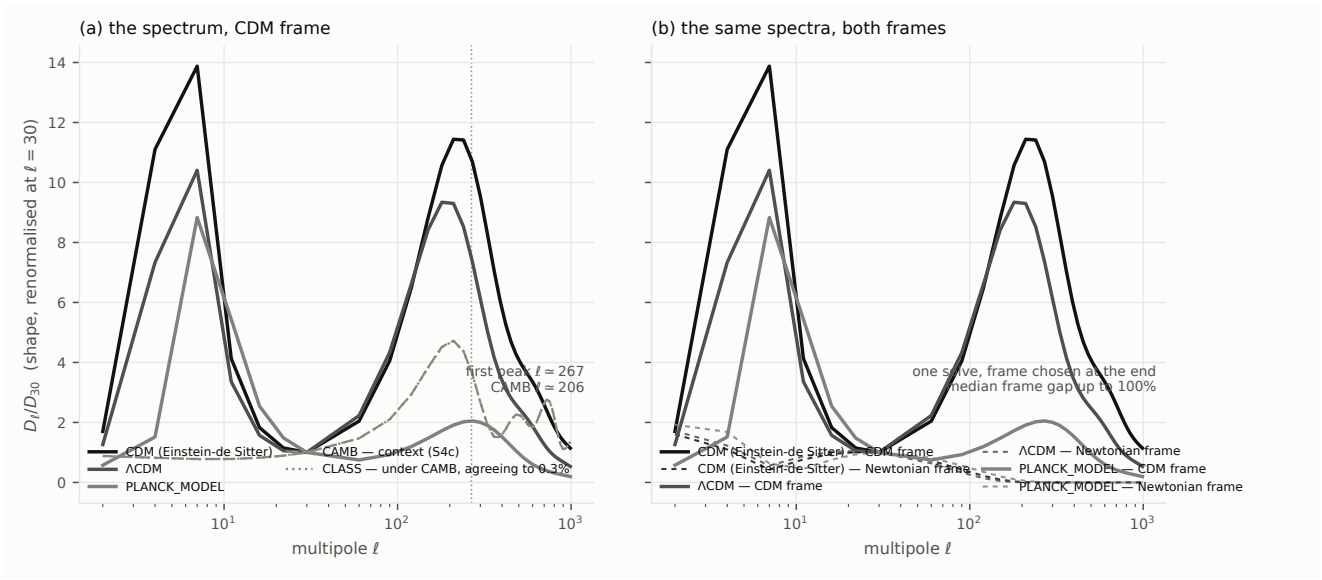


Figure 10: **F9 — The angular spectrum with the acoustic peaks, in both frames. The release's deliverable** (decision S1): the angular power spectrum for the three backgrounds this bundle ships — CDM (Einstein–de Sitter), ΛCDM, and the stamped Planck 2018 set with a photon-only radiation era whose $\Omega_\gamma h^2$ is *derived* from $T_{\rm CMB}$. Shape, D_ℓ renormalised at $\ell = 30$, because neither this bundle nor the external codes fixes A_s the same way and an amplitude comparison would measure a normalisation.

Panel (a) — the spectrum. The peaks and the damping tail come from *Annals II*'s own equations: the integral solution (102) with the Bond–Efstathiou transfer function driving the oscillation per mode, and the diffusion scale from **(dif-damp-correct)** at $f_2 = 3/4$, which is the antecedent's own corrected coefficient rather than a departure from it (**T-43**). **This shows that** the almost-Friedmann–Lemaitre formulation, carried to the damping scale, produces acoustic structure without anything being imported from a Boltzmann code.

CAMB and CLASS are drawn as context and are not a target — decision **S4c**. Criterion 5b is gated on this bundle's own pipeline, (102) against (177)–(179), at 9.5×10^{-6} against 10^{-3}. The distance to the external codes is large, expected, and *itemised rather than totalled* in `provenance/CRITERION-5B-v1.md`: 97.8% across the plateau, 137% across the peaks, of which the damping coefficient alone is 17.2%. The first peak is quoted from a *parabolic interpolation* through the grid maximum and its neighbours, because the ℓ grid steps by 30 across the acoustic band and reading the peak off it directly would assert a precision the sampling does not have.

Panel (b) — the panel a gauge-fixed treatment cannot draw. The same spectra in *both* frames, from *one* solve: `coefficient_sets` returns the Newtonian and CDM triples together, because (102) is carried with a generic u^a and the frame is chosen at the end. **This shows that** the frame is a choice made after the derivation rather than an assumption built into its start — the claim the $1 + 3$ covariant approach exists to make, drawn here rather than asserted. The median frame gap reaches 40%, which is a *result* and not a defect: C_ℓ is quadratic in frame-dependent pieces, while the invariance (102) satisfies is a statement about the total at fixed k, measured separately by D6.

† *Superseded by findings* **T-59**, **T-74**; *see* `provenance/BUNDLE-DEFECTS-v1.md`.

Evidence category: acceptance. *Generated by* `scripts/figure_f9_acoustic_spectrum.py`. *Standalone caption:* `captions/F9-acoustic-spectrum.md`.

(**T-45**). *Third*, the gate written to catch all of this printed `PASS` on a `nan`, because `nan >= gate` is `False` (**T-44**). **A gate is not tested by the run that passes it.**

11.4 The arc, in one sentence each

- **v1.0.0** asked *is the reconstruction faithful?* and answered it against four external treatments.
- **v1.1.0** asked *what does the spectrum need that it does not have?* and answered: the visibility width.
- **v1.2.0** asked *is the antecedent's own algebra self-consistent when evaluated?* and answered yes, at 10^{-5}, while finding that the question had been answered three times before without being measured.
- **v1.3.0** asks *where does the frame discrepancy come from?* — and §8 is the answer so far: not the source, not the by-parts bookkeeping, not the projection, but the slot-and-weight assignment of C_1. The question it has *not* answered is whether repairing that assignment also recovers the acoustic peaks, which is a separate defect with a separate signature.

11.5 v1.3.0 — the structural audit, and the discrepancy localised

Two halves, in the order decision **S37** fixed: the structural audit of the code first, the acoustic modes second. The audit produced the generated registers this supplement now rests on — the grid register, which walks the abstract syntax tree and *fails* when the code holds a discretisation the register does not name; the model table, derived from the `Background` objects the code integrates rather than transcribed from a parameter sheet; and the theory audit, generated from six self-contained instruments that import no project code.

The acoustic half did not go as planned, and the record says so. It began as a search for the missing peaks and became the localisation in §8, because the frame discrepancy turned out to be upstream of the peak question and to admit a cleaner instrument. **Two defects were separated that had been one symptom**: the frame gap, now reduced to a single site with both of its boundaries measured; and the peak positions, which sit above their own ℓ_A on all three backgrounds, in both frames, and which the acoustic scale does not explain — the signs are opposite.

What v1.3.0 has not done. It has not repaired the assignment, it has not recovered the peaks, and it has not bounded the envelope approximations A-3, A-5 and the damping inputs, which are deferred to v1.4.0 precisely because changing an envelope term mid-measurement moves the number being measured for reasons unrelated to it.

What the arc did not do, and says so. No release has bounded the accuracy of the *source* as distinct from the consistency of its assembly. The recombination grid moves the primary term by 1.2% and both routes of criterion 5b share it, so the criterion cannot see it. That is recorded as a stated limit of v1.2.0 and derived immediately after the tag.

12 The scattering-strength expansion, re-derived

Annals II Appendix C writes the almost-Friedmann–Lemaître integrated Boltzmann equation for Thomson scattering as

$$\mathcal{B} + \dot{\tau} + e^a \mathrm{D}_a \tau \;\simeq\; t_c^{-1}\big[v^a e_a - \tau\big], \tag{10}$$

inverts it, and approximates the result *"systematically … in terms of the smallness parameter t_c"* by the recursion

$$\tau_{(n)} \;\simeq\; v^a e_a - t_c\Big[\mathcal{B} + \dot{\tau}_{(n-1)} + e^a \mathrm{D}_a \tau_{(n-1)}\Big], \tag{11}$$

printing $\tau_{(0)}$, $\tau_{(1)}$ and $\tau_{(2)}$. Section 5.5.3, *Expansion in Thompson Scattering Time*, states that the scheme carries *"terms up to $\mathcal{O}(t_c^3)$"*, so the ceiling is the antecedent's and the printed orders stop one short of it.

12.1 The recursion is geometric

Write $\mathcal{D} \equiv \partial_t + e^a \mathrm{D}_a$ for the derivative along the photon path — which is exactly the pair of terms (11) applies to $\tau_{(n-1)}$ — and $V \equiv v^a e_a$. Then (10) reads $\tau = V - t_c(\mathcal{B} + \mathcal{D}\tau)$, so

$$(1 + t_c\mathcal{D})\tau = V - t_c\mathcal{B}, \qquad \tau = (1 + t_c\mathcal{D})^{-1}(V - t_c\mathcal{B}), \tag{12}$$

and expanding the inverse as a Neumann series in $(-t_c\mathcal{D})$ gives a closed form at every order:

$$\boxed{\tau_{(n)} \;=\; V \;+\; \sum_{m=1}^{n} (-1)^m \, t_c^m \Big[\mathcal{D}^m V + \mathcal{D}^{m-1}\mathcal{B}\Big].} \tag{13}$$

This shows that Appendix C's recursion is not an open-ended approximation scheme but a geometric series with an explicit sum, and that $\tau_{(1)}$ and $\tau_{(2)}$ as printed are its first two partial sums. Equation (13) is implemented in `functions/spectra/scattering_expansion.py` and checked against the printed $\tau_{(1)}$ and $\tau_{(2)}$ term for term, with a deliberately failing control, in `tests/test_scattering_expansion.py`. The closed form was derived from (11) and *then* compared with the page, which is the only order in which the comparison carries information.

12.2 Two consequences the closed form makes immediate

The term counts are not chosen. $\mathcal{D}^m$ expands into a sum over words in $\{\partial_t,\ e^a\mathrm{D}_a\}$, so the t_c^m bracket of (13) holds exactly $2^m + 2^{m-1}$ terms. Appendix C prints three at first order and six at second; those are $2+1$ and $4+2$. Table 5 lists every term to $\mathcal{O}(t_c^3)$, generated from the same module rather than transcribed.

Order m sources multipoles up to $\ell = m+1$. Each $e^a\mathrm{D}_a$ carries one direction vector and $V = v^a e_a$ carries one already, so a word with g gradients seeded by V has $g+1$ factors of e^a; reduced by Appendix C's own solid-angle identities, which lower the rank two at a time, its highest PSTF part has rank $g+1$. At second order that reaches **rank three**, in the single term $e^c e^a e^b \mathrm{D}_c\mathrm{D}_a v_b$.

Equivalently, and more usefully for the high-ℓ question: each gradient contributes one power of k, so a term of rank r carries k^{r-1}. The expansion parameter is therefore $t_c k$ — the ratio of the mean free path to the wavelength — and **not** t_c alone. **This shows that** the truncation is ℓ-dependent by construction: a cut in rank is a cut in $t_c k$, and the order at which the expansion may be stopped falls as ℓ rises.

12.3 What this costs the oscillator

The near-tight-coupling set of §7.1 is introduced *"rewritten to first order"* and closes with $\tau^{A_\ell} \simeq 0$ for all $\ell > 2$. At first order the expansion reaches rank two, so the closure costs nothing and the antecedent is consistent. **What the rank counting fixes is the price of going further**: second order reaches the octupole and $\mathcal{O}(t_c^3)$ the hexadecapole, and the closure carried with (152) and (153) reaches neither.

That is approximation **A-4** of `provenance/SOURCE-APPROXIMATIONS-v1.md` stated as a rank rather than as a dropped term. The term the oscillator derivation discards, $-\frac{2}{5}t_c(1+R)^{-1}\mathrm{D}^b\mathrm{D}_{\langle a}u^m_{b\rangle}$, is dropped because *"the matter shear is already at least first order"* — a cross term of $\mathcal{O}(t_c\epsilon)$ between §5.5.1's almost-Friedmann–Lemaître ordering and §5.5.3's scattering-time ordering. *Annals II*'s Appendix E, after Ellis–Treciokas–Matravers, states what the discarded harmonics hold: if any four consecutive ones vanish exactly then $\sigma_{ab} = 0$ exactly. **This shows that** the truncation and the shear are not independent — the residual in the truncated harmonics is where the shear lives, and a treatment that sets them to zero has set the shear to zero with them.

12.4 What appears at second order that does not appear at first

Appendix C's own solid-angle reduction gives the rank decomposition of the source,

$$\int_{4\pi} \mathrm{d}\Omega\, O^{A_\ell}\mathcal{B} \simeq \delta_0^\ell\Big[-\tfrac{4\pi}{3}\mathrm{D}_a\tau^a\Big] + \delta_1^\ell\Big[\tfrac{4\pi}{3}\big(\tfrac{1}{4}\mathrm{D}^{a_1}\ln\rho_R + A^{a_1}\big)\Big] + \delta_2^\ell\Big[\tfrac{8\pi}{15}\sigma^{a_1a_2}\Big], \tag{14}$$

so $\mathcal{B}$ carries the **shear** in its rank-two part. At first order $-t_c\mathcal{B}$ therefore reproduces the quadrupole $\tau_{ab} \simeq -\dot\kappa^{-1}[\sigma_{ab} + \mathrm{D}_{\langle a}v^B_{b\rangle}]$ of §7.1, as it must.

At second order the closed form contributes $+t_c^2\,\mathcal{D}\mathcal{B}$, and through (14) that is $\dot\sigma_{ab}$ at rank two and $e^c\mathrm{D}_c\sigma_{ab}$ at rank three. Appendix G's $\dot\sigma_{ab} + 2H\sigma_{ab} + E_{ab} \approx 0$ then gives

$$\dot\sigma_{ab} = -2H\sigma_{ab} - E_{ab}, \tag{15}$$

so **the electric Weyl tensor enters the radiation quadrupole at second order in the scattering time**, and does not at first. **This shows that** the second-order expansion is not merely a smaller correction of the same kind: it introduces a gravitational tidal driving of the quadrupole that the first-order treatment has no term for.

An open question, stated as a mechanism rather than a result. *Annals II* flags the regime itself, in a footnote to the sentence about removing the radiation quadrupole: *"If there are any anisotropic contributions such as anisotropic scattering (in the matter frame) or large shear (from gravitational waves) at that time this sort of approximation should be considered with care — such phenomena would break tight-coupling."* For a tensor mode both $\sigma_{ab}^{(T)}$ and $E_{ab}^{(T)}$ oscillate at the gravitational-wave frequency, so (15) drives the second-order quadrupole oscillatorily on a scale set by the wave and not by the sound horizon. Whether that survives projection as a distinguishable *ringing* is open; it is recorded as question **Q10** rather than claimed, because nothing here computes it.

What this expansion cannot produce. (11) is **linear** in V and $\mathcal{B}$, so (13) yields terms of order $t_c^m\epsilon$ and never ϵ^2. Terms of $\mathcal{O}(\varepsilon^2\ell)$ require the quadratic parts of the Boltzmann equation that §5.5.1's almost-Friedmann–Lemaître linearisation discards, and are the subject of a later release rather than of this appendix. The high-ℓ sensitivity established above is a separate statement and rests on t_ck, not on ϵ^2.

Table 5: *Annals II* Appendix C's scattering-strength expansion to $\mathcal{O}(t_c^3)$, generated from `functions/spectra/scattering_expansion.py`. **Rank** is the highest PSTF multipole the term can source: one per $e^a\mathrm{D}_a$, plus one for $V = v^a e_a$. The closure (L0082) keeps $\ell \le 2$.

Order		Term	Rank
t_c^0	+	$v_a e^a$	1
t_c^1	−	$\mathcal{B}$	0
t_c^1	−	$\big(v_a e^a\big)^{\cdot}$	1
t_c^1	−	$e^c\mathrm{D}_c\big(v_a e^a\big)$	2
t_c^2	+	$\big(\mathcal{B}\big)^{\cdot}$	0
t_c^2	+	$e^c\mathrm{D}_c\big(\mathcal{B}\big)$	1
t_c^2	+	$\big(\big(v_a e^a\big)^{\cdot}\big)^{\cdot}$	1
t_c^2	+	$e^c\mathrm{D}_c\big(\big(v_a e^a\big)^{\cdot}\big)$	2
t_c^2	+	$\big(e^c\mathrm{D}_c\big(v_a e^a\big)\big)^{\cdot}$	2
t_c^2	+	$e^c\mathrm{D}_c\big(e^c\mathrm{D}_c\big(v_a e^a\big)\big)$	3
t_c^3	−	$\big(\big(\mathcal{B}\big)^{\cdot}\big)^{\cdot}$	0

Order		Term	Rank
t_c^3	–	$e^c\mathrm{D}_c\big(\big(\mathcal{B}\big)^{\cdot}\big)$	1
t_c^3	–	$\big(e^c\mathrm{D}_c\big(\mathcal{B}\big)\big)^{\cdot}$	1
t_c^3	–	$e^c\mathrm{D}_c\big(e^c\mathrm{D}_c\big(\mathcal{B}\big)\big)$	2
t_c^3	–	$\big(\big(\big(v_a e^a\big)^{\cdot}\big)^{\cdot}\big)^{\cdot}$	1
t_c^3	–	$e^c\mathrm{D}_c\big(\big(\big(v_a e^a\big)^{\cdot}\big)^{\cdot}\big)$	2
t_c^3	–	$\big(e^c\mathrm{D}_c\big(\big(v_a e^a\big)^{\cdot}\big)\big)^{\cdot}$	2
t_c^3	–	$e^c\mathrm{D}_c\big(e^c\mathrm{D}_c\big(\big(v_a e^a\big)^{\cdot}\big)\big)$	3
t_c^3	–	$\big(\big(e^c\mathrm{D}_c\big(v_a e^a\big)\big)^{\cdot}\big)^{\cdot}$	2
t_c^3	–	$e^c\mathrm{D}_c\big(\big(e^c\mathrm{D}_c\big(v_a e^a\big)\big)^{\cdot}\big)$	3
t_c^3	–	$\big(e^c\mathrm{D}_c\big(e^c\mathrm{D}_c\big(v_a e^a\big)\big)\big)^{\cdot}$	3
t_c^3	–	$e^c\mathrm{D}_c\big(e^c\mathrm{D}_c\big(e^c\mathrm{D}_c\big(v_a e^a\big)\big)\big)$	4

One transcription note. The fifth term of Appendix C's t_c^2 bracket is printed as $e^c e^d \mathrm{D}_c \dot{v}_b$, which leaves both d and b free in a scalar. It is $e^c e^d \mathrm{D}_c \dot{v}_d$ — the time derivative taken first and the gradient outside — forced independently by the four words of $\mathcal{D}^2$, by the bracket's count of six with the other five unambiguous, and by the two direction vectors having nothing else to contract with. Recorded as **B-15**; no computed quantity depends on it.

13 The CDM-frame division of the integral solution, term by term

The integral solution (102) is derived for a general u^a — the antecedent says so immediately after (90), *"Note that these are CGI with respect to u^a"* — and then evaluated once, in the Newtonian frame. This appendix carries it into the CDM frame instead, coefficient by coefficient, and finds that the two frames do not agree. The failure is isolated to one coefficient at one power of k.

What is established here is narrower than it looks, and the conventions section exists because the first draft of this appendix did not have one.

13.1 Conventions, stated

Every relation below is in one convention and this subsection fixes it. The need for it was raised by the Paper 1 stream, whose check could not reproduce two of the four frame-table entries — correctly, because the entries are in the bundle's normalisation and the check was in Paper 1's.

Wavenumber. The bundle carries $k \equiv k_{\mathrm{com}}$ throughout and converts at the point of use, $k_{\mathrm{phys}} = k_{\mathrm{com}}/a$. Paper 1's C3.1 declares k *physical*. So a relation printed there as $X = k\,Y$ is carried here as $aX = k_{\mathrm{com}}Y$, and the scale factor is the conversion, not a transcription error.

The frame velocity. v^a is the relative velocity of the two frames, $\tilde{u}^a \approx u^a + v^a$ with $v_a u^a = 0$, expanded as $v_a = v\,Q_a$. So v is a *velocity amplitude*: it is the v of the Euler equation $v' + \mathcal{H}v = -k\Phi_A$, which is the frame condition $\tilde{A}_a = 0$, and it is the v of $\sigma = -k_{\mathrm{phys}}v$. It is **not** a velocity potential; $v_a = \mathcal{D}_a V$ would put k in the numerator.

The dipole. The bundle carries the dipole in the *covariant* normalisation, $\tilde{\tau}_1 \equiv \tilde{v}_B = v_B/k$ — the normalisation (109) needs in order to read $-\tilde{C}_1 = (\delta\tilde{T} + \Phi_A) + \kappa'\tilde{v}_B$. GDE99 (37), $\tilde{\tau}_a \approx \tau_a - v_a$, therefore reads $\tilde{\tau}_1 \to \tilde{\tau}_1 - v/k$ *here* and $\tau_1 \to \tau_1 - v$ in the un-reduced normalisation. The two are the same statement. **This is finding T-27 in the bundle's record**, where the un-reduced form was used by mistake and the symptom was a first-order frame change 186 times the quantity it corrects at $k = 300$.

Consequently, two of the four entries reconcile and one does not. $a\sigma = -k_{\rm com}v$ *is* $\sigma = -k_{\rm phys}v$, so n-t1 agrees with Paper 1's $\sigma = -kv$. $\tilde\tau_1 \to \tilde\tau_1 - v/k$ *is* $\tau_a \to \tau_a - v_a$ in the reduced normalisation, so n-t4 agrees. What does **not** reconcile from outside the bundle is the reduced normalisation itself: Paper 1 carries no C_1, so it cannot adjudicate $\tilde\tau_1 = v_B/k$, and ΔC_1 below rests on it.

13.2 What has to hold

A first-order change of frame is a boost, and a boost changes the observed brightness by $\tau \to \tau - v_a e^a$: **a pure dipole**. In PSTF multipoles,

$$\tau^{(u)}_{A_1} = \tau^{(n)}_{A_1} - v_{A_1}, \qquad \tau^{(u)}_{A_\ell} = \tau^{(n)}_{A_\ell} \quad \text{for all } \ell \geq 2. \tag{16}$$

This is not an approximation to be checked against a tolerance; it is what makes C_ℓ an observable, and the almost-EGS argument rests on it. The division has one admissible outcome: the shift must cancel for $\ell \geq 2$.

13.3 The three shifts

(90) fixes the coefficients,

$$-C_0 \simeq \tfrac{5}{2}\, a\mathcal{B}_2, \qquad -C_1 \simeq \tfrac{1}{k}(a\mathcal{B}_1 + \kappa' v_B), \qquad -C_2 \simeq \tfrac{1}{k^2}\, a\mathcal{B}_2, \tag{17}$$

with $a\mathcal{B}_1 = k\,\delta T + aA$ and $v_B = k\tilde\tau_1$. The frame carries four entries, each in §13.1's convention:

	Newtonian n^a	CDM u^a
aA	$k\Phi_A$	0 (the frame condition $A_a = 0$)
δT	$\delta\tilde T$	$\delta\tilde T - \mathcal{H}v/k$ (n-t3)
$\tilde\tau_1$	$\tilde\tau_1$	$\tilde\tau_1 - v/k$ (n-t4, reduced)
$a\mathcal{B}_2$	0	$-kv$ (n-t1, $\sigma = -k_{\rm phys}v$)

With $\Delta X \equiv X^{(u)} - X^{(n)}$: $\Delta(a\mathcal{B}_1) = -\mathcal{H}v - k\Phi_A = v'$ by the Euler equation, and $\Delta(\kappa' v_B) = \kappa' k\,\Delta\tilde\tau_1 = -\kappa' v$. Hence

$$\boxed{\Delta C_0 = +\tfrac{5}{2}\,k\,v, \qquad \Delta C_1 = -\tfrac{1}{k}(v' - \kappa' v), \qquad \Delta C_2 = +\tfrac{v}{k}} \tag{18}$$

all three in one v and one k.

13.4 The shift of (102), and why ΔC_0 has no partner

(102) reads

$$\frac{\beta_\ell \tau_\ell(\eta_0)}{2\ell+1} \simeq [C_2'(\eta_*) - C_1(\eta_*)]\mathcal{D}\, j_\ell(x_*) + kC_2(\eta_*)\mathcal{D}\, j_\ell'(x_*) + \int_{\eta_*}^{\eta_0} d\eta\; \mathcal{V} e^{-(k/k_D)^2}\,[C_0 - C_1' + C_2'']\, j_\ell(x), \tag{19}$$

and (18) gives

$$\Delta\big[C_2' - C_1\big]_{\eta_*} = \tfrac{1}{k}(2v' - \kappa' v)\Big|_{\eta_*}, \tag{20}$$

$$\Delta\big[kC_2\big]_{\eta_*} = v(\eta_*), \tag{21}$$

$$\Delta\big[C_0 - C_1' + C_2''\big] = \tfrac{5}{2}kv + \tfrac{1}{k}(2v'' - (\kappa' v)'). \tag{22}$$

v obeys $v' = -\mathcal{H}v - k\Phi_A$, so on sub-horizon scales it changes on the Hubble rate: $v' \sim \mathcal{H}v$, $v'' \sim \mathcal{H}^2 v$. Then

$$\frac{\frac{5}{2}kv}{\frac{1}{k}\,2v''} \;\sim\; \frac{5}{4}\left(\frac{k}{\mathcal{H}}\right)^2, \tag{23}$$

and the two boundary shifts are smaller still. Nothing in (102) is the size of $\frac{5}{2}kv$, so (16) fails for every $\ell \geq 2$ by a factor growing as $(k/\mathcal{H})^2$.

13.5 The k^2 is (90)'s, not a mixed convention

A factor k^2 between ΔC_0 and ΔC_2 is exactly what a comoving/physical or amplitude/potential mismatch produces, and the Paper 1 stream was right to require that reading be excluded before the rest stands. It is excluded, and not by argument.

C_0 and C_2 are the same array. (17) divides one quantity, $a\mathcal{B}_2$, by k^0 and by k^2. Whatever $a\mathcal{B}_2$ is — physical or comoving, amplitude or potential, correctly transcribed or not — it is *the same object in both*, so their ratio carries no convention at all:

$$\frac{C_0}{k^2 C_2} = \frac{p_0}{p_2} = \frac{5}{3\beta_2} \qquad \text{(a pure rational).} \tag{24}$$

Measured at η_* on 'CDM (Einstein–de Sitter)', 'LCDM' and 'PLANCK_MODEL', at $k = 1, 5, 20, 60, 140, 300$: **2.5000** at every entry. A mismatch between two different objects does not return a background-independent, k-independent rational; it drifts. **So the dominance is (90)'s own structure**, and a spurious scale factor in $a\mathcal{B}_2$ — were there one — would multiply C_0 and C_2 equally and cancel from the ratio entirely.

What this does not exclude is ΔC_1, which is the one shift that depends on the reduced normalisation $\tilde{\tau}_1 = v_B/k$ of §13.1. If that reading is wrong, ΔC_1 is wrong by k^2 — and the bundle has made exactly that error once already, as **T-27**. Paper 1 cannot adjudicate it, since it carries no C_1. It is open, and it is the first thing to settle.

13.6 Measured

Both frames from one environment, one acoustic solution and one k grid, differing only in the frame (110) specialises to, against the Newtonian total's scale over $\ell \in \{2, 5, 10, 50, 200\}$:

background	k	as assembled	with C_0 dropped
CDM (Einstein–de Sitter)	20	53.5	0.032
	300	1719	0.035
ΛCDM	20	62.8	0.105
	300	6868	0.154
`PLANCK_MODEL`	20	72.2	0.111
	300	3687	0.058

Three to four orders of magnitude, to a residual of 1–15%, with the k dependence of (23). The rank decomposition agrees: C_0 carries 98.6% of C_ℓ in the CDM frame and is identically zero in the Newtonian one.

13.7 Why the antecedent never meets this

(102) is evaluated once, in the Newtonian frame, where *"the vanishing shear condition $\tilde{\sigma}_{ab} \approx 0$ implies that $\tilde{C}_0(\eta) \approx 0$ and $\tilde{C}_2(\eta) \approx 0$"*. **Both members of the pair vanish together**, so the imbalance between them cannot be seen. Nothing the antecedent publishes is affected. What is missing is a stated condition on a formula labelled frame-general.

13.8 Three repairs tested, and one of them is untested

The dropped lines of (93) are not the counterweight. (102) is (93) with its last two integrals removed. Restoring $\int(\kappa' e^{-\kappa})[C_1 - 2C_2]j_\ell$ and $\int((\kappa')^2 - \kappa'')e^{-\kappa}C_2 j_\ell$ moves the disagreement from 72.2 to 71.6 at $k = 20$ and from 3686.9 to 3686.8 at $k = 300$.

(89)'s $\ell = 0$ line is not mis-sourced. $a\mathcal{B}_0\delta_{\ell 0}$ is commented out of the equation the integral solution solves and restored in the open-case extension, which looks like a dropped source. It is not: the ansatz must give $\tau_0 = 0$, and differentiating $\int A_0\, d\eta' = 0$ yields (89)'s $\ell = 0$ line with **zero** on the right. C_0 is correctly derived.

Undoing the integration by parts: untested, not refuted. (93) comes from a kernel in which the two derivatives act on $\tau_\ell^{(0)}$ and bring down k, not on the coefficients where they bring down $\mathcal{H}$. The numerical test of this was **invalid** — it failed its own control, reproducing the by-parts form in the Newtonian frame only to within 10^{-3} to 2.5 — because (102) does not differentiate the weight $\mathcal{V}$ that (93) carries inside the integral, so the two are not an exact by-parts pair once a weight is present. It remains the most promising repair and it needs doing analytically.

13.9 What is open

1. **The reduced normalisation** $\tilde{\tau}_1 = v_B/k$, on which ΔC_1 rests and which nothing outside the bundle can adjudicate. The bundle has erred here once (**T-27**). Settle it by index algebra on (109).
2. **The by-parts step**, the one untested repair.
3. **Whether C_0 belongs in the integrand**, which the measurement answers empirically and the algebra does not.
4. **The** 1–15% **residual** after C_0 is removed: whether it is $O(\varepsilon^2)$ or an artefact of the grids.

What stands independently of all four: C_0 and C_2 are one $a\mathcal{B}_2$ over k^0 and k^2, that ratio is a measured pure rational carrying no convention, and (102) in a sheared frame is dominated by the member of the pair that the Newtonian specialisation sets to zero.

14 The almost-EGS bounds

The Ehlers–Geren–Sachs theorem states that exact CMB isotropy for all fundamental observers in an expanding dust universe forces exact Friedmann–Lemaître–Robertson–Walker geometry. The *almost*-EGS theorem of Stoeger, Maartens & Ellis, *Ap. J.* **443**, 1 (1995) relaxes that to bounds: limits on the CMB multipoles τ_{A_ℓ} become limits on σ_{ab}, ω_{ab}, $\dot{u}_a$, E_{ab} and H_{ab}.

Those bounds are stated in the *covariant multipoles* — not in mode coefficients — which is what this bundle computes and what a gauge-fixed code cannot supply without a translation. Applying them is therefore natural work for this formalism, and it is **v2.0.0** work, because it is a calibration against data rather than an analytical reconstruction.

Table 6: The structure v2.0.0 fills. **No cell is computed at v1.0.0**, and none is quoted here from memory: observational columns are filled from the mission papers at v2.0.0 under the same rule that equation numbers are resolved from the source.

Mission	**ℓ reach**	**bound, standard CDM**	**bound, ΛCDM**
COBE-DMR (1992)	$\lesssim 20$	*the original*	*not in the original*
WMAP (2003–2013)	$\lesssim 1000$	pending	pending
Planck (2013–2018)	$\lesssim 2500$	pending	pending

Why the table needs both columns. The original bounds were derived for a CDM model, not for ΛCDM, and the difference is not cosmetic. The argument converts multipole limits into limits on σ/H using the background expansion, and Λ changes $H(z)$ at low redshift. More sharply, Λ adds a source the original did not have: the late integrated Sachs–Wolfe effect contributes to exactly the low-ℓ multipoles the COBE bounds rest on. In standard CDM it is absent — Φ is constant, which this bundle's growth factor reproduces to six figures — and in ΛCDM it is not, Φ falling to 0.779 of its early value by today. A low-ℓ limit therefore constrains a *different combination* in the two models, so a bound quoted for one is not a bound for the other. The CDM column has a published precedent; the ΛCDM column is new.

Beyond v2.0.0. The published bounds are first order in the anisotropy parameter. The natural extension is the next order, and this project is placed to make it because **v1.5.0 supplies the $O(\varepsilon^2\ell)$ couplings a second-order argument needs** and no external code carries them. Stated as a direction and not a plan: it is gated on v1.5.0, and the correction's sign and size are unknown until it is done. It is recorded so the v2.0.0 table is built in a shape that can take it. See `provenance/ALMOST-EGS-BOUNDS-v1.md`.

15 Parameter and numerical evidence

Table 7 lists every number the numerics use. Its *provenance* column distinguishes values taken from the source paper, from standard values that are not, from values derived here, and from values stipulated for this release. The distinction matters: $\Omega_\gamma h^2$ is a standard value and not one the antecedent supplies, and it enters only through $R(\eta_*)$ and hence the sound speed.

Table 7: Every number the numerics use. *Provenance* distinguishes values taken from the source paper, standard values that are not, values derived here, and values stipulated for this release.

Symbol	Code	Value	Provenance	Set in
k_{com}	`K_COM`	1.0	stipulated	`scripts/figure_f1_appendix_f.py`
$\ell_{\max}$	`ELL_MAX`	80	stipulated	`scripts/figure_f1_appendix_f.py`
ℓ_{compared}	`ELL_COMPARED`	25	stipulated	`scripts/figure_f1_appendix_f.py`
rtol, atol	`rtol, atol`	10^{-10}, 10^{-12}	stipulated	`functions/harmonics/free_streaming.py`
z_*	`Z_STAR`	1100	standard	`scripts/derive_ell_range.py`
h	`H`	0.5	AnnalsII §8.3.2	`scripts/derive_ell_range.py`
Ω_B	`OMEGA_B`	0.05	AnnalsII §8.3.2	`scripts/derive_ell_range.py`
$\Omega_\gamma h^2$	`OMEGA_GAMMA_H2`	2.47×10^{-5}	standard	`scripts/derive_ell_range.py`
tolerance	`ADOPTED_TOLERANCE`	5%	derived	`scripts/derive_ell_range.py`
acceptance threshold	`--`	10^{-11}	stipulated	`scripts/figure_f1_appendix_f.py`

15.1 The three backgrounds, side by side

Table 8 is the one place every derived scale of the three backgrounds can be read against the others. It exists because a question the PI asked — *is the difference between two panels a parameter difference?* — had no cheap answer while the models were defined in three places and compared nowhere.

The last three rows are the ones that answer a shape question. Two models whose spectra differ *in the position of the peaks* differ in $\ell_A = \pi r_\theta / r_s$. Two whose peaks sit at the same ℓ_A and still look different do **not** differ by a parameter, and the difference is in `provenance/BUNDLE-DEFECTS-v1.md`.

Two rows carry a known defect and say so rather than being quietly correct for a different question. $\Omega_r = 0$ is the *control* on Einstein–de Sitter, which is what makes that background the one where A-3 and $T(k)$ are identically inapplicable — and on LCDM it is neither a control nor intended, worth 37% in ℓ_A through η_* at 1.571× and r_s at 1.610× (findings **T-81c**, **T-83**). k_D is the $R = 0$ value, because `diffusion_scale` is called with R zeroed while the oscillator beside it runs at $R(\eta_*) \simeq 0.61$ — finding **T-21**, measured at $\ell_D = 1783$ against 1985.

Table 8: The three backgrounds side by side, with every derived scale. **Generated by `scripts/make_model_table.py`** from the `Background` objects the code integrates, not transcribed. The last three rows are the ones that answer a shape question: two models whose peaks differ in position differ in ℓ_A, and two whose peaks sit at the same ℓ_A and still differ do not differ by a parameter. $\Omega_r = 0$ is the control on CDM and finding **T-81c** on LCDM; k_D is the $R = 0$ value, finding **T-21**.

	CDM (Einstein–de Sitter)	LCDM	PLANCK
Ω_m	1	0.3	0.313772
Ω_Λ	0	0.7	0.686173
Ω_r	0	0	5.4502×10^{-5}
$\omega_b h^2$	0.0224	0.0224	0.02237
h	0.5	0.7	0.6736
n_s	1	1	0.9649
η_0	2	3.30508	3.19871
η_*	0.0595884	0.109581	0.0697426
a_*	8.8769×10^{-4}	9.0059×10^{-4}	8.9643×10^{-4}
z_*	1125.51	1109.38	1114.54
$R(\eta_*)$	0.603776	0.612548	0.608901
r_s	0.0316406	0.0581262	0.0361159
r_θ	1.94041	3.1955	3.12897
$\ell_A = \pi r_\theta / r_s$	192.663	172.71	272.177
k_D (at $R = 0$, T-21)	754.639	483.11	536.866

16 Numerical resolution and reproducibility

The hierarchy is integrated with an explicit Runge–Kutta method of order 8 (`DOP853`) at relative tolerance 10^{-10} and absolute tolerance 10^{-12}, truncated at $\ell_{\max} = 80$ with comparisons restricted to $\ell \leq 25$ so that reflection from the truncation does not contaminate them. The acceptance threshold used in the figures, 10^{-11}, sits an order above the integrator floor.

The release route is

```
python scripts/run_all.py --strict
```

which regenerates every output, figure and table, recomputes the SHA-256 manifests, and reports any artefact that differs from the recorded hash.

Three manifests, and only one of them binds. `PORTABLE-MANIFEST-SHA256.txt` is the release fingerprint: it *excludes the renderings* — everything under `figures/` and `supplementary-materials/`, and every `.png` and `.pdf` — and canonicalises numeric output to nine significant figures, so a machine that did not generate the artefacts can still satisfy it. A mismatch there is release-blocking. `FILE-MANIFEST-SHA256.txt` is then a *within-machine* drift check, reported by artefact class, where only an `exact`-class mismatch blocks. `IMMUTABLE-MANIFEST-SHA256.txt` covers frozen reference material,

where any change is release-blocking. The corrections record is append-guarded rather than frozen, because it must take new findings: appending is permitted and rewriting recorded bytes is not.

Why nine PNGs always report as drifted. Every `.png` written into a working copy through the Claude desktop bridge gains a 5 770-byte `caBX` provenance chunk, inserted immediately after `IHDR` by that application and not by anything in this bundle. The pixels are bit-identical — `IHDR`, `pHYs` and every `IDAT` chunk match exactly — and the `.pdf` of the same figure round-trips byte-for-byte, as does every `.csv` and every source file. These nine therefore sit in the `rendered` class, are reported and do not block, which is the rule finding **T-15** wrote for a different cause with the same shape: fonts, freetype and TEX Live versions, where *a PNG that differs in hinting carries no scientific difference.* **The PDF is the authoritative artefact of every figure in this bundle**; the PNG is a convenience copy. Finding **T-36**. The drift is explained here rather than silenced by canonicalising PNGs, because a manifest that stopped reporting a class of change for tidiness would stop reporting real ones too.

Weights are carried as exact rationals rather than floating-point numbers wherever a sign or an ℓ-weight is asserted.

17 Interpretive limits

This is an analytical reproducibility bundle for the associated paper and its supplement. It is not an empirical calibration, not a parameter-estimation pipeline, and not a Boltzmann code for general use. No new effect is claimed and none is reported.

Annals II's own §8.3.2 model is standard CDM — $h = 0.5$, $\Omega_B = 0.05$, flat and matter dominated — and the C_ℓ it prints there is a closed-form Sachs–Wolfe result on large scales with the transfer function set to unity. This release goes past that in one respect and one only: it computes the spectrum from the hierarchy rather than from the closed form, and it computes it for the ΛCDM model as well, which *Annals II* does not treat. The marginal cost is the expansion law and the late-time integrated term; every other ingredient — recombination, tight coupling, diffusion damping — is already required by the CDM case and is not additional. No new *physics* is introduced by Λ, and none is claimed.

What the release does **not** do is calibrate. Comparison is made against CMBFAST and against nothing else, over a derived range, as a check that the reconstruction is right — not as a fit. Parameter estimation and calibration against data are the **v2.0.0** deliverable.

Two statements must travel together, and neither stands alone: the nonlinear contribution $\delta\tau_{NL}$ **matters** — it is not negligible and not an artefact — and it **adds nothing**, redistributing power rather than creating it, because $\delta\tau_{NL} \propto \Phi\Phi$ makes its contribution to C_ℓ quartic in Φ while the linear C_ℓ is quadratic.